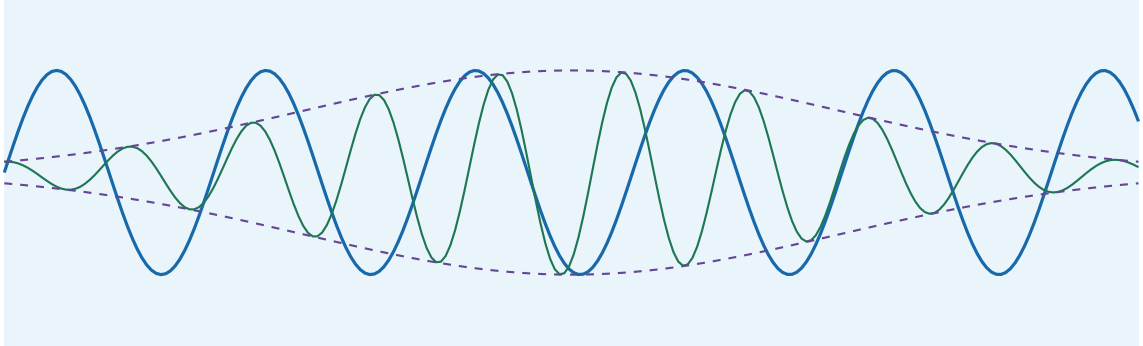
WAVES

Complete Learning Notes

Oscillators, mechanical waves, and light

Version V4.1 October 19, 2026

Organized by Shang Wu



Definition: Learning sequence

Restore equilibrium → diagonalize coupled motion → take the continuum limit → propagate energy → connect light to matter.

Expanded derivations, standardized solution routes, physical checks, and a visual mind map at the end of every numbered chapter. V4.0 content is retained in full.

Based on the supplied Waves syllabus and the relevant sections of Peatross and Ware, *Physics of Light and Optics*, May 15, 2024 revision. All instructional prose, worked examples, and figures in this edition are newly prepared.

V4.1 Coverage Audit and Additions

Key Result: Coverage is present; explanatory depth is strengthened

The eight units and explicit topic bullets of the supplied Waves syllabus were already represented in V4.0. This revision retains that content and adds derivation workshops, explicit decision steps, extra diagrams, and ten chapter-ending mind maps. Coverage of a syllabus does not establish coverage of unavailable lecture details or predict an examination.

Unit	Explicit syllabus items audited	V4.1 strengthening
1	Free motion, phase and initial conditions; local equilibrium; averaged energy and power; damping and Q; forced amplitude/phase at low, high and resonant frequency; phasors, linearity and superposition	Characteristic-root solution with initial data; phase-space ellipse; response phase graph; power and energy diagnostics; stepwise switched-drive template.
2	Longitudinal springs; transverse springs, small-amplitude and slinky limits; two masses and three springs; normal modes and beating	Mass-metric modal proof; force projection and reconstruction; exact beat-energy caution; transverse spring expansion; antiresonance example.
3	Many-coordinate modes and dispersion; periodic boundaries; solid-state/sound connection; binary chain and phonons; forced motion	Discrete orthogonality; acoustic slope derivation; band-edge eigenvectors; equal-mass folding; physical unit-cell diagram; zero-mode forcing.
4	String wave equation; transverse standing modes; boundary conditions; wavenumber/wavelength and dispersion; superposition and Fourier analysis	Boundary-based orthogonality proof; complete triangular-pluck integration; modal-energy normalization; Fourier partial-sum illustration; data-compatibility checklist.
5	Forward/backward traveling waves; phase/group velocity; energy flux and impedance; superposition; transmission/reflection from a barrier	Force/velocity sign convention; complex junction response; finite-layer matching route; energy checks; reflection/transmission graph and barrier geometry.
6	Maxwell to wave equation; plane waves and polarization; dispersion, phase speed, impedance and superposition; Poynting theorem, stored energy and power; absorption	All four Fourier-space Maxwell constraints; physical current accounting; complex impedance versus real power; standing electric/magnetic energy plot; standard field-reconstruction route.
7	Lorentz oscillator model; real/imaginary susceptibility; connection to absorption and phase velocity	Exact complex square-root steps; microscopic energy and transparent dispersive stored energy; approximation decision tree; index/absorption graph and inverse-response example.
8	Plane-wave interference; polarization effects and visibility; unequal frequencies and group velocity; standing waves	Polarization-ellipse axes; visibility inversion and coherence qualification; detector averaging derivation; group delay and packet assumptions; contrast graph.

How to use the expanded chapters

Each chapter first presents the original learning sequence, followed by a derivation and problem-solving workshop. Numbered routes specify the recognition trigger, assumptions, algebra or projection, final output, and an independent check. Each numbered chapter then ends with a four-branch mind map connecting the model, calculation method, physical meaning, and error checks. The matching V4.0 practice volume remains applicable; its problem IDs are referenced throughout.

The explicit source boundary remains unchanged: Berkeley mechanics volumes and Prof. Moti Segev's lecture notes named by the syllabus were not supplied. No new school-specific examination claims are made. Topics outside the explicit syllabus remain labeled as supplementary or as optional mathematical tools. All additional figures and examples are newly prepared.

Course Map and Reading Guide

Definition: What this edition covers

The required scope is the supplied two-page *Waves – Syllabus*: eight units from single oscillators through electromagnetic interference. Peatross and Ware provide the optical reference, not a replacement syllabus. Their ray optics, imaging, diffraction, holography, and blackbody chapters are not included as compulsory topics here.

Unit	Required coverage	Location and practice
1	Free, damped, and forced oscillators; phase; initial data; power; quality factor; phasors	Chapter 2; Problems O1–O8. Mathematical tools in Chapter 1.
2	Longitudinal and transverse spring coupling; slinky limit; two masses/three springs; modes and beats	Chapter 3; C1–C8.
3	Many degrees of freedom; dispersion; periodic boundaries; monatomic and binary lattices; forced response	Chapter 4; L1–L8.
4	String wave equation; transverse standing modes; boundary conditions; Fourier superposition	Chapter 5; S1–S8.
5	Traveling waves; phase and group speeds; energy flux; impedance; reflection and transmission	Chapter 6; T1–T8.
6	Maxwell to wave equation; plane waves; polarization; impedance; Poynting theorem; absorption	Chapter 7 and polarization in Chapter 9; E1–E8 and I2–I3.
7	Lorentz oscillator; complex susceptibility; absorption and phase speed in a medium	Chapter 8; M1–M8.
8	Interference; polarization overlap; visibility; unequal frequencies; group velocity; electromagnetic standing waves	Chapter 9; I1–I8.

How the supplied textbook is used

Page numbers below are the book's printed pages, not the PDF viewer counter (the main text has an eight-page offset).

- Chapter 0, Sections 0.1–0.4 (pp. 1–18): complex numbers, vector calculus, linear algebra, and Fourier tools.
- Sections 1.6–1.7 (pp. 34–38): polarization sources and the electromagnetic wave equation.
- Sections 2.1–2.4, 2.6–2.7 (pp. 43–51, 54–58): plane waves, complex refractive index, Lorentz model, energy transport. Section 2.5 (pp. 52–53) is used only for a labeled conductor/plasma extension.
- Chapter 6, Sections 6.1–6.6 (pp. 144–152): polarization notation and optional Jones-matrix tools.

- Chapter 7, Sections 7.1–7.5 (pp. 170–182): interference, beats, spectra, packets, and dispersion. A full coherence-theory chapter is not assumed to be part of the required syllabus.

The syllabus names Berkeley volumes I and III for Units 1–5 and Prof. Moti Segev's lecture notes for electromagnetism. These materials were not supplied. The mechanical chapters therefore provide self-contained derivations aligned to the syllabus; they are not presented as a page-by-page reconstruction of unavailable lectures. No recent course examinations or tutorials were supplied, so no school-specific frequency ranking is asserted.

Standard Workflow: A route through the subject

For each model, identify the coordinates and approximations, derive the equation, solve its eigenvalue or response problem, impose the data or boundaries, and finish with an energy or limiting-case check. The same sequence reappears from two masses to a dielectric.

Color key

Blue: definitions and assumptions. Green: governing results and complete solutions. Purple: worked examples. Gold: standard workflows. Red: error checks. Gray: supplementary material beyond the explicit syllabus wording.

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1 Tools, Conventions, and the Physical Strategy

Source map: Syllabus Units 1–8; Peatross and Ware Sections 0.1–0.4.

1.1 From a physical model to a solvable equation

An oscillation is repeated motion near equilibrium; a wave is organized motion distributed over space. Their common mathematical core is a linear restoring mechanism. A single displacement obeys an ordinary differential equation. A set of displacements gives a matrix equation. A continuum gives a partial differential equation. A sinusoidal trial solution converts each into an algebraic problem.

Definition: The linear regime

Expand a smooth potential near a stable equilibrium q_0:

$$U(q_0+\eta) = U(q_0) + U'(q_0)\eta + \tfrac{1}{2}U''(q_0)\eta^2 + O(\eta^3).$$

At equilibrium $U'(q_0)=0$; stability requires $U''(q_0)>0$ in the nondegenerate one-coordinate case. Neglecting higher orders gives $m\ddot{\eta} + U''(q_0)\eta = 0$. This is a local approximation, not a statement that every large-amplitude motion is harmonic.

1.2 One phasor convention throughout

Write a real harmonic quantity as

$$q(t) = \mathrm{Re}\{Q e^{-i\omega t}\}, \qquad Q = Ae^{i\phi}, \qquad A \geq 0.$$

Then $q = A\cos(\omega t - \phi)$, $\dot{q} \leftrightarrow -i\omega Q$, and $\ddot{q} \leftrightarrow -\omega^2 Q$. A positive argument of Q means a positive time lag relative to a cosine drive. For a forward spatial wave use $e^{i(kx-\omega t)}$; its complex amplitude includes e^{ikx}.

Key Result: Average a product, not the factors separately

If $a = \mathrm{Re}(Ae^{-i\omega t})$ and $b = \mathrm{Re}(Be^{-i\omega t})$, then

$$\langle ab \rangle = \tfrac{1}{2}\,\mathrm{Re}(AB^*).$$

Proof: insert $a = (Ae^{-i\omega t} + A^* e^{i\omega t})/2$, multiply, and discard the terms at 2ω. In particular $\langle a^2 \rangle = |A|^2/2$, whereas $\langle a \rangle = 0$.

Worked Example: Phase and power from a complex amplitude

Let $F(t) = 6\cos(4t)\,\mathrm{N}$ and $X = (0.03 + 0.04i)\,\mathrm{m}$. Then $|X| = 0.05\,\mathrm{m}$ and $\phi = \mathrm{atan2}(0.04, 0.03) = 53.13°$. Thus $x = 0.05\cos(4t - 53.13°)$. The velocity phasor is $V = -4iX = 0.16 - 0.12i\,\mathrm{m\,s^{-1}}$, so

$$\langle F\dot{x} \rangle = \tfrac{1}{2}\,\mathrm{Re}(6V^*) = 0.48\,\mathrm{W}.$$

Multiplying the real parts of the two phasors would not give the required average.

1.3 Linear algebra and mass-weighted orthogonality

For conservative coordinates $\boldsymbol{q}$, let M be a symmetric positive-definite mass matrix and K the symmetric stiffness matrix. A mode $\boldsymbol{q} = \boldsymbol{a}\cos\omega t$ satisfies

$$K\boldsymbol{a} = \omega^2 M\boldsymbol{a}.$$

Two eigenvectors with different ω^2 obey $\boldsymbol{a}_r^T M \boldsymbol{a}_s = 0$: transpose the two eigenvalue equations and subtract, using symmetry. The natural inner product is therefore $\langle \boldsymbol{u}, \boldsymbol{v} \rangle_M = \boldsymbol{u}^T M \boldsymbol{v}$, not always the

ordinary dot product. With $\boldsymbol{z} = M^{1/2}\boldsymbol{q}$, the matrix $D = M^{-1/2}KM^{-1/2}$ is symmetric and diagonalizable by an orthogonal matrix.

Standard Workflow: Modal initial-value problems

Find the eigenpairs, normalize $\boldsymbol{a}_r^T M\boldsymbol{a}_s = \delta_{rs}$, project $Q_r(0) = \boldsymbol{a}_r^T M\boldsymbol{q}(0)$ and $\dot{Q}_r(0) = \boldsymbol{a}_r^T M\dot{\boldsymbol{q}}(0)$, then reconstruct $\boldsymbol{q} = \sum_r \boldsymbol{a}_r Q_r(t)$. A degeneracy allows a choice of orthogonal basis within the degenerate subspace; it does not remove modes.

1.4 Fourier and vector-calculus essentials

Orthogonality replaces coupled equations by independent coefficients. On $0 < x < L$,

$$\int_0^L \sin\frac{n\pi x}{L}\sin\frac{m\pi x}{L}\,\mathrm{d}x = \frac{L}{2}\delta_{nm}.$$

Our spatial Fourier transform pair is

$$\widehat{f}(k) = \int_{-\infty}^{\infty} f(x)\mathrm{e}^{-\mathrm{i}kx}\mathrm{d}x, \qquad f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \widehat{f}(k)\mathrm{e}^{\mathrm{i}kx}\mathrm{d}k.$$

For a plane wave, $\nabla \to \mathrm{i}\boldsymbol{k}$, $\partial_t \to -\mathrm{i}\omega$. Two identities needed later are

$$\nabla\times(\nabla\times\boldsymbol{E}) = \nabla(\nabla\cdot\boldsymbol{E}) - \nabla^2\boldsymbol{E},$$
$$\nabla\cdot(\boldsymbol{E}\times\boldsymbol{H}) = \boldsymbol{H}\cdot(\nabla\times\boldsymbol{E}) - \boldsymbol{E}\cdot(\nabla\times\boldsymbol{H}).$$

Error Check: Units and amplitudes

ω is in radians per second, while $f = \omega/(2\pi)$ is in hertz. Here $k = 2\pi/\lambda$ is angular wavenumber. RMS amplitude is peak amplitude divided by $\sqrt{2}$ for a scalar sinusoid. Using an RMS field in a formula already containing $1/2$ loses another factor of two.

Worked Example: Scaling without solving

If a string under tension $\mathcal{T}$ has linear density μ, the only speed made from them is $v = \sqrt{\mathcal{T}/\mu}$: $[\mathcal{T}] = \mathrm{kg\,m\,s^{-2}}$ and $[\mu] = \mathrm{kg\,m^{-1}}$. Dimensional analysis predicts this combination, but not the boundary-dependent factor $n\pi/L$ in a standing-mode frequency. Use dimensions as a check, not as a substitute for boundary conditions.

1.5 Derivation Workshop: Why the Standard Tools Work

1.5.1 A complete mass-orthogonality proof

Assume $M = M^T > 0$ and $K = K^T$. From $K\boldsymbol{a}_r = \omega_r^2 M\boldsymbol{a}_r$, premultiply by $\boldsymbol{a}_s^T$ to get $\boldsymbol{a}_s^T K\boldsymbol{a}_r = \omega_r^2\boldsymbol{a}_s^T M\boldsymbol{a}_r$. Transpose the equation for s and multiply by $\boldsymbol{a}_r$:

$$\boldsymbol{a}_s^T K\boldsymbol{a}_r = \omega_s^2\boldsymbol{a}_s^T M\boldsymbol{a}_r.$$

Subtracting gives $(\omega_r^2 - \omega_s^2)\boldsymbol{a}_s^T M\boldsymbol{a}_r = 0$. Distinct eigenvalues therefore imply orthogonality in the mass metric. For a repeated eigenvalue this argument yields $0 = 0$; apply Gram–Schmidt using $\langle\boldsymbol{u},\boldsymbol{v}\rangle_M = \boldsymbol{u}^T M\boldsymbol{v}$ inside the eigenspace instead.

To see completeness, set $D = M^{-1/2}KM^{-1/2}$, which is real symmetric. Choose an orthonormal eigenbasis $\boldsymbol{b}_r$ of D and define $\boldsymbol{a}_r = M^{-1/2}\boldsymbol{b}_r$. Then $\boldsymbol{a}_r^T M\boldsymbol{a}_s = \delta_{rs}$ and the square matrix $A = (\boldsymbol{a}_1,\ldots,\boldsymbol{a}_N)$ satisfies $A^T MA = I$, hence $A^{-1} = A^T M$. This last identity proves the reconstruction and projection formulas, rather than just motivating them.

Worked Example: Recover a modal amplitude with the correct normalization

Let $M = \mathrm{diag}(2, 1)$ in mass units and let an unnormalized mode be $\boldsymbol{a} = (1, 2)^T$. For initial position $\boldsymbol{q}_0 = (3, 0)^T$, its coefficient in a mass-orthogonal modal expansion is

$$c = \frac{\boldsymbol{a}^T M \boldsymbol{q}_0}{\boldsymbol{a}^T M \boldsymbol{a}} = \frac{6}{2+4} = 1.$$

Ordinary Euclidean projection would give $3/5$, a different number. If $\widehat{\boldsymbol{a}} = \boldsymbol{a}/\sqrt{6}$ is used instead, the normalized coordinate is $Q_0 = \sqrt{6}$; the physical contribution $\widehat{\boldsymbol{a}} Q_0 = \boldsymbol{a}$ is unchanged.

1.5.2 Orthogonality of Fourier sines, with all cases separated

Use $2 \sin\alpha \sin\beta = \cos(\alpha - \beta) - \cos(\alpha + \beta)$. For $n \neq m$, each cosine integrates to a sine evaluated at an integer multiple of π, giving zero. For $n = m$, do not divide by $n - m$; use $\sin^2\alpha = (1 - \cos 2\alpha)/2$ instead:

$$\int_0^L \sin^2(n\pi x/L)\mathrm{d}x = L/2.$$

This is the origin of the coefficient $2/L$. Similarly, $\int_0^L \cos^2(n\pi x/L)\mathrm{d}x = L/2$ for positive integer n, but the constant cosine mode has integral L. Zero modes need separate normalization.

Standard Workflow: The model-to-answer route

1. **Recognize the object:** one coordinate, a finite vector, or a function on a domain.
2. **State the approximation:** small displacement, small slope, linear response, or narrow bandwidth are different assumptions.
3. **Choose the representation:** characteristic roots for transients; one phasor per frequency for a drive; eigenvectors for coupled coordinates; Fourier functions for an initial profile.
4. **Project or impose data:** use the inner product and boundary conditions belonging to that representation.
5. **Reconstruct the real observable:** take the real part of a linear field before interpreting motion, and use a conjugated average for power.
6. **Check independently:** units, one initial condition, one symmetry, and one limiting case are inexpensive tests.

1.6 Chapter Mind Map

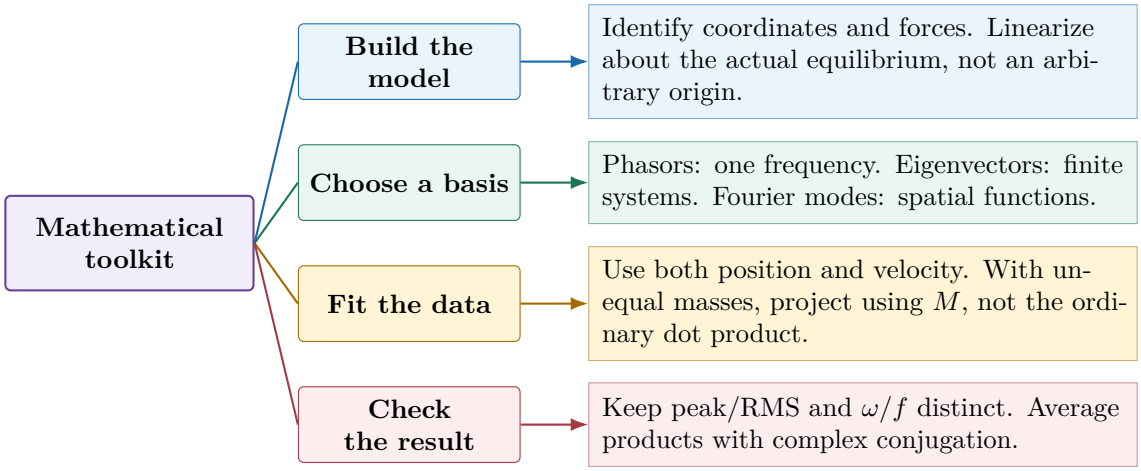


Figure 1.1: Chapter 1 mind map: representation is chosen by the structure of the physical problem.

2 Single Oscillators: Motion, Response, and Energy

Source map: Syllabus Unit 1; self-contained mechanical derivations; phasor connection to Peatross and Ware Section 0.2.

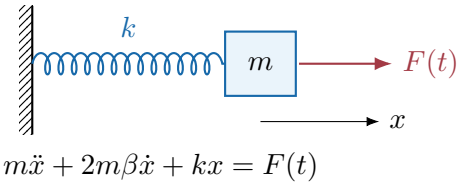


Figure 2.1: A linear oscillator measured from equilibrium. The damping force is $-2m\beta\dot{x}$.

2.1 Free motion and initial conditions

Newton's law is $m\ddot{x} = -kx$, so $\omega_0 = \sqrt{k/m}$ and

$$x(t) = C\cos\omega_0 t + D\sin\omega_0 t = x_0\cos\omega_0 t + \frac{v_0}{\omega_0}\sin\omega_0 t.$$

Equivalently $x = A\cos(\omega_0 t - \phi)$, with $A = \sqrt{x_0^2 + (v_0/\omega_0)^2}$ and $\phi = \operatorname{atan2}(v_0/\omega_0, x_0)$. The phase is necessary to encode both initial position and velocity; a plain $A\cos\omega_0 t$ cannot fit arbitrary data.

Key Result: Energy of a free oscillator

$$E = \tfrac{1}{2}m\dot{x}^2 + \tfrac{1}{2}kx^2 = \tfrac{1}{2}kA^2, \qquad \langle T\rangle = \langle U\rangle = \frac{E}{2}.$$

Multiplying $m\ddot{x} + kx = 0$ by $\dot{x}$ gives $\dot{E} = 0$. The instantaneous kinetic and potential energies oscillate at $2\omega_0$ while their sum stays constant.

Worked Example: From a nonlinear force to a local oscillator

For $U(x) = ax^4 - bx^2$ with $a, b > 0$, equilibria satisfy $2x(2ax^2 - b) = 0$. At $x = 0$, $U'' = -2b < 0$, so small disturbances grow. At $x_* = \pm\sqrt{b/(2a)}$, $U'' = 12ax_*^2 - 2b = 4b$. Hence $\eta = x - x_*$ oscillates with $\omega_0 = 2\sqrt{b/m}$ for $|\eta| \ll |x_*|$. Stable equilibrium, rather than a globally quadratic potential, is the reason harmonic motion appears.

2.2 Damping: three regimes and an exact initial-value solution

With a velocity-proportional resistance,

$$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = 0, \qquad s^2 + 2\beta s + \omega_0^2 = 0.$$

Key Result: The three damping cases

$$\begin{aligned} 0 \le \beta < \omega_0: &\quad x = \mathrm{e}^{-\beta t}(C\cos\omega_d t + D\sin\omega_d t), \quad \omega_d = \sqrt{\omega_0^2 - \beta^2}, \\ \beta = \omega_0: &\quad x = (C + Dt)\mathrm{e}^{-\omega_0 t}, \\ \beta > \omega_0: &\quad x = C\mathrm{e}^{s_+ t} + D\mathrm{e}^{s_- t}, \quad s_\pm = -\beta \pm \sqrt{\beta^2 - \omega_0^2}. \end{aligned}$$

For underdamping, $C = x_0$ and $D = (v_0 + \beta x_0)/\omega_d$. The extra βx_0 follows by differentiating the exponential envelope; omitting it violates the initial velocity.

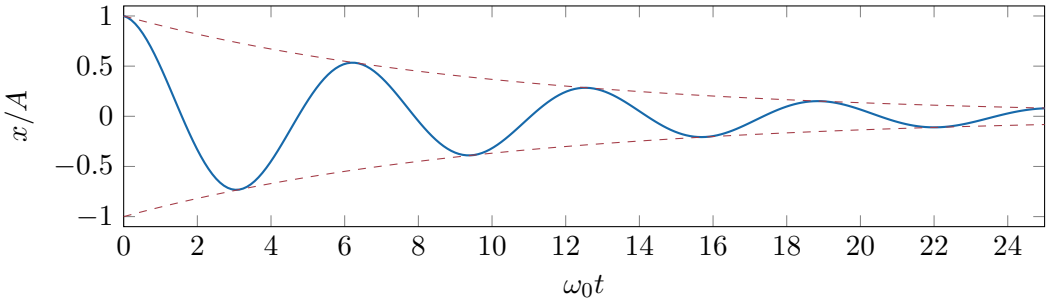


Figure 2.2: Underdamped motion and its displacement envelope. The energy envelope decays twice as fast as amplitude in weak damping.

The exact mechanical energy obeys

$$\dot{E} = m\dot{x}(\ddot{x} + \omega_0^2 x) = -2m\beta\dot{x}^2 \le 0.$$

Writing $x = A\mathrm{e}^{-\beta t}\cos\theta$, $\theta = \omega_d t - \phi$, gives the exact expression

$$E = \tfrac{1}{2}mA^2\mathrm{e}^{-2\beta t}\{(\beta\cos\theta + \omega_d\sin\theta)^2 + \omega_0^2\cos^2\theta\}.$$

The braces are not constant. For $\beta \ll \omega_0$, a local cycle average changes little over one cycle and gives $\overline{E} \simeq \tfrac{1}{2}m\omega_0^2 A^2\mathrm{e}^{-2\beta t}$. Thus $\tau_A = 1/\beta$, $\tau_E = 1/(2\beta)$, and the weak-damping loss factor is $Q \simeq \omega_0/(2\beta)$. We use $Q = \omega_0/(2\beta)$ as the standard oscillator parameter even outside the regime where a sharp resonance exists.

Worked Example: Infer damping from ring-down data

An amplitude falls to 1/e of its initial envelope in 5.0 s; the measured oscillation frequency is $\omega_d = 10.0\,\mathrm{s}^{-1}$. Then $\beta = 0.200\,\mathrm{s}^{-1}$, $\omega_0 = \sqrt{100 + .0400} = 10.002\,\mathrm{s}^{-1}$, and $Q = 25.005$. The energy envelope reaches 1/e in 2.5 s. If successive same-sign peaks have ratio R, their logarithmic decrement is $\delta = \ln R = 2\pi\beta/\omega_d$ and $Q = \tfrac{1}{2}\sqrt{1 + (2\pi/\delta)^2} \simeq \pi/\delta$.

2.3 Forced response: amplitude and phase are one complex number

For a drive $F_0\cos\omega t$, take $x_{\mathrm{ss}} = \mathrm{Re}(X\mathrm{e}^{-\mathrm{i}\omega t})$. Substitution gives

$$m(\omega_0^2 - \omega^2 - 2\mathrm{i}\beta\omega)X = F_0.$$

Key Result: Transfer function and phase lag

$$X = \frac{F_0/m}{\omega_0^2 - \omega^2 - 2\mathrm{i}\beta\omega}, \qquad A = \frac{F_0/m}{\sqrt{(\omega_0^2-\omega^2)^2 + 4\beta^2\omega^2}},$$

$$\phi = \operatorname{atan2}(2\beta\omega, \omega_0^2 - \omega^2), \qquad x_{\mathrm{ss}} = A\cos(\omega t - \phi).$$

The full solution is $x = x_{\mathrm{ss}} + x_{\mathrm{transient}}$; only the second term can impose the remaining initial data. A damped transient decays, leaving a response at the drive frequency, not at ω_d.

Regime	Amplitude	Lag
$\omega \ll \omega_0$	$A \simeq F_0/k$	$\phi \simeq 0$
$\omega = \omega_0$	$A = F_0/(2m\beta\omega_0)$	$\phi = \pi/2$
$\omega \gg \omega_0, \beta$	$A \simeq F_0/(m\omega^2)$	$\phi \to \pi$

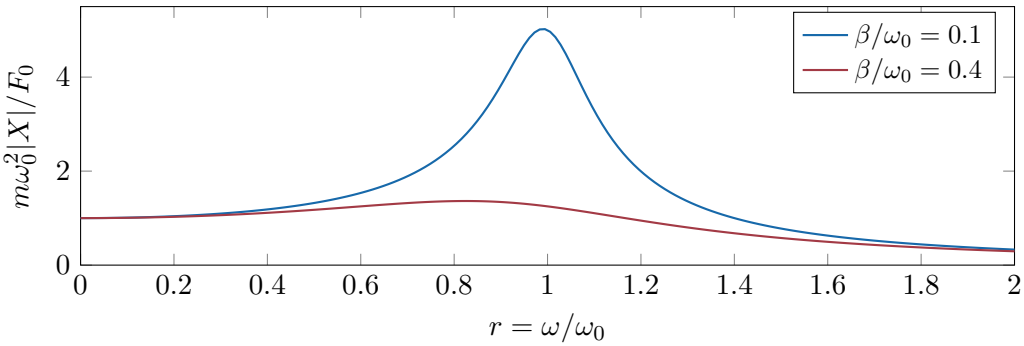


Figure 2.3: Displacement response for two damping strengths. Increased damping broadens and lowers the response.

2.4 Which resonance is being asked for?

Minimize $D(\omega) = (\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2$. Since $D' = 4\omega(\omega^2 - \omega_0^2 + 2\beta^2)$, a nonzero displacement maximum exists only if $\beta < \omega_0/\sqrt{2}$, at

$$\omega_{\mathrm{amp}} = \sqrt{\omega_0^2 - 2\beta^2}.$$

For stronger damping, the displacement amplitude decreases from its static value; there is no nonzero-frequency displacement peak. This does not eliminate a peak in absorbed power.

Key Result: Mean absorbed power and energy balance

$$\overline{P} = \langle F\dot{x}\rangle = \frac{1}{2}F_0 A\omega\sin\phi = m\beta\omega^2 A^2 = \frac{\beta F_0^2\omega^2/m}{(\omega_0^2-\omega^2)^2 + 4\beta^2\omega^2}.$$

For frequency-independent viscous damping, $\overline{P}$ is maximal at $\omega = \omega_0$, with $P_{\max} = F_0^2/(4m\beta)$. The velocity amplitude peaks there too. The steady average energy satisfies input power = dissipated power.

At half power, divide the denominator by ω^2 to obtain $(\omega_0^2/\omega - \omega)^2 = 4\beta^2$. Thus

$$\omega_- = \sqrt{\omega_0^2 + \beta^2} - \beta, \quad \omega_+ = \sqrt{\omega_0^2 + \beta^2} + \beta, \quad \Delta\omega = 2\beta, \quad Q = \frac{\omega_0}{\Delta\omega}.$$

These are exact for the power curve of this model. The approximate displacement linewidth is the same only in the narrow-resonance limit.

Worked Example: Compare three frequencies in one experiment

Take $m = 1\,\mathrm{kg}$, $\omega_0 = 5\,\mathrm{s}^{-1}$, $\beta = 1\,\mathrm{s}^{-1}$, $F_0 = 2\,\mathrm{N}$. Free ring-down occurs at $\sqrt{24} = 4.899\,\mathrm{s}^{-1}$; the displacement resonance is at $\sqrt{23} = 4.796\,\mathrm{s}^{-1}$; maximum power is at $5\,\mathrm{s}^{-1}$. At the last frequency, $A = 0.2\,\mathrm{m}$, $\phi = 90°$, and $\overline{P} = 1\,\mathrm{W}$. The half-power frequencies are $\sqrt{26} \mp 1$, whose difference is exactly $2\,\mathrm{s}^{-1}$.

2.5 Linearity, resonance without damping, and a standard answer

If $F = \sum_j F_j \cos(\omega_j t + \alpha_j)$, add the individual responses. Do not combine different frequencies into one phasor. At zero damping and exact resonance, a bounded steady-state ansatz fails: for a drive turned on at rest,

$$x(t) = \frac{F_0}{2m\omega_0} t \sin \omega_0 t.$$

Substitution verifies the equation and initial data; the growing envelope explains why dividing by $\omega_0^2 - \omega^2$ at resonance is invalid.

Standard Workflow: Single-oscillator questions

Locate the equilibrium and linearize; divide by m to identify ω_0 and β; separate free/transient motion from driven steady motion; determine both amplitude and phase; apply initial data to the full solution; distinguish the requested resonance; check units and power loss. If damping is zero, treat exact resonance separately before dividing.

Error Check: A free oscillator does not have a frequency-response peak by itself

Its natural frequency is a property of the homogeneous motion. Resonance is a statement about response to a drive or energy transfer. In weak damping ω_d, ω_{amp}, and ω_0 are close, but they are not interchangeable in an exact derivation.

2.6 Derivation Workshop: Initial Data and Measurable Responses

2.6.1 Derive all three initial-value solutions from the roots

For $\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = 0$, the exponential trial yields $s_\pm = -\beta \pm \sqrt{\beta^2 - \omega_0^2}$. In underdamping the roots are $-\beta \pm \mathrm{i}\omega_d$; taking real independent combinations gives the damped sine and cosine. At $t = 0$,

$$x(0) = C = x_0, \qquad \dot{x}(0) = -\beta C + \omega_d D = v_0,$$

so $D = (v_0 + \beta x_0)/\omega_d$. At critical damping, the repeated-root second solution is $t\mathrm{e}^{-\omega_0 t}$, and the data give

$$x(t) = [x_0 + (v_0 + \omega_0 x_0)t]\mathrm{e}^{-\omega_0 t}.$$

In overdamping, set $d = \sqrt{\beta^2 - \omega_0^2} > 0$. Solving $C_+ + C_- = x_0$ and $s_+ C_+ + s_- C_- = v_0$ gives

$$C_+ = \frac{v_0 - s_- x_0}{2d}, \qquad C_- = \frac{s_+ x_0 - v_0}{2d}.$$

The apparent singularity at $d = 0$ means the two exponential basis functions have become dependent. It is resolved by the critical-damping form, not by assigning infinite physical motion.

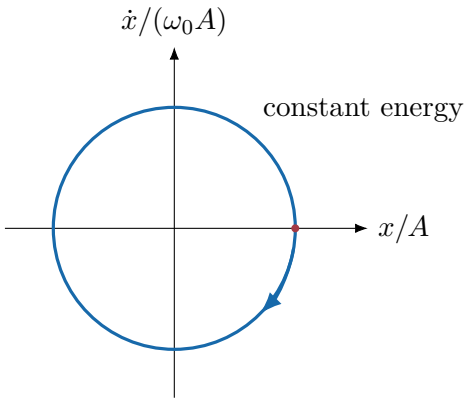


Figure 2.4: For free undamped motion, $x^2/A^2 + \dot{x}^2/(\omega_0 A)^2 = 1$. The trajectory is clockwise: at maximum positive displacement the acceleration points toward the origin.

2.6.2 Phase space makes the initial condition visible

The equation above follows by squaring $x = A\cos(\omega_0 t - \phi)$ and $\dot{x} = -\omega_0 A \sin(\omega_0 t - \phi)$ and adding. One point (x_0, v_0) determines the energy ellipse and the phase on it. With damping, energy decreases and the trajectory spirals inward in the underdamped case; critical and overdamped motions approach the origin without repeated rotations. This distinguishes a position zero crossing from a change in the total energy.

2.6.3 Response phase and the physical meaning of its three limits

The force balance for a steady drive is the vector sum of an elastic contribution kx, a damping contribution $2m\beta\dot{x}$, and an inertial contribution $m\ddot{x}$. At low frequency inertia and damping are small, so force nearly follows displacement. At ω_0, the elastic and inertial phasors cancel, leaving force proportional to velocity; displacement therefore lags by $\pi/2$. At high frequency inertia dominates, so acceleration follows force and displacement is almost opposite to it.

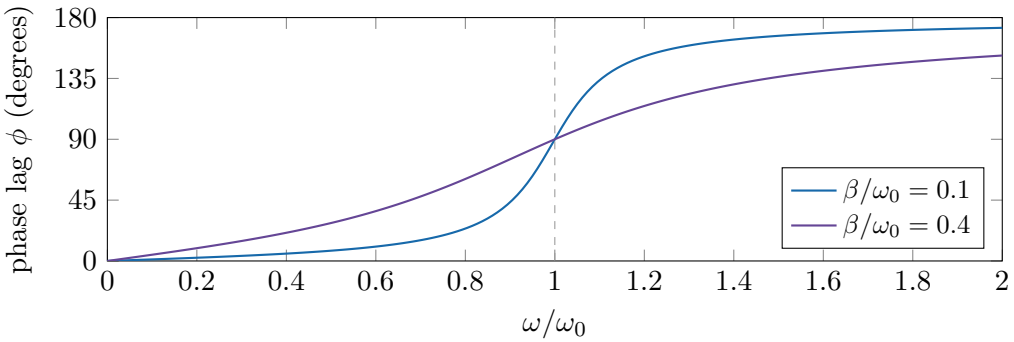


Figure 2.5: Every positive damping strength gives a 90° lag at ω_0, even when a sharp displacement peak is absent.

2.6.4 Quality factor: parameter, linewidth, and energy-loss meanings

For a decaying underdamped motion, the exact same-phase energy ratio over $T_d = 2\pi/\omega_d$ is $E(t + T_d)/E(t) = \mathrm{e}^{-2\beta T_d}$. The exact fraction lost is $1 - \mathrm{e}^{-2\beta T_d}$. Therefore the quantity $2\pi E/\Delta E$ computed from a finite free-decay cycle is $2\pi/[1 - \mathrm{e}^{-4\pi\beta/\omega_d}]$, not exactly $\omega_0/(2\beta)$. They agree to leading order for weak damping. The familiar ring-down energy interpretation of Q is thus an approximation; the parameter $Q = \omega_0/(2\beta)$ and the power-linewidth identity remain unambiguous for this model.

The average mechanical energy in a driven steady sinusoid is exact:

$$\overline{E}_{\mathrm{ss}} = \frac{mA^2}{4}(\omega^2 + \omega_0^2).$$

At $\omega = \omega_0$, $\overline{E}_{\mathrm{ss}} = m\omega_0^2 A^2/2$ and $\overline{P} = m\beta\omega_0^2 A^2$, so $\omega_0 \overline{E}_{\mathrm{ss}}/\overline{P} = Q$. Off resonance the ratio changes. State which experiment and definition are being used.

Worked Example: Complete a switched-drive solution without guessing its phase

Suppose a unit-mass oscillator satisfies $\ddot{x} + 2\dot{x} + 10x = 6\cos 2t$ and begins at $x = \dot{x} = 0$. The steady phasor is $X = 6/(6 - 4\mathrm{i}) = (9 + 6\mathrm{i})/13$. Hence $x_{\mathrm{ss}} = (9\cos 2t + 6\sin 2t)/13$. Since $\omega_d = 3$, write $x = x_{\mathrm{ss}} + \mathrm{e}^{-t}(C\cos 3t + D\sin 3t)$. The position gives $C = -9/13$, and the velocity gives $12/13 - C + 3D = 0$, hence $D = -7/13$. Thus

$$x(t) = \frac{9\cos 2t + 6\sin 2t - \mathrm{e}^{-t}(9\cos 3t + 7\sin 3t)}{13}.$$

Both initial conditions vanish; the transient decays while the drive frequency remains.

Standard Workflow: Standard routes for oscillator questions

Given initial data: classify the roots → write two independent real solutions → solve two data equations → check the velocity by differentiation.
Given a sustained harmonic force: write the complex dynamic stiffness → invert it → obtain modulus and `atan2` phase → calculate the requested displacement, velocity, energy, or power peak separately.
Given measured decay or linewidth: identify whether the measurement is amplitude, energy, hertz, or angular frequency → infer β and ω_0 → calculate Q with the relevant approximation stated.

2.7 Chapter Mind Map

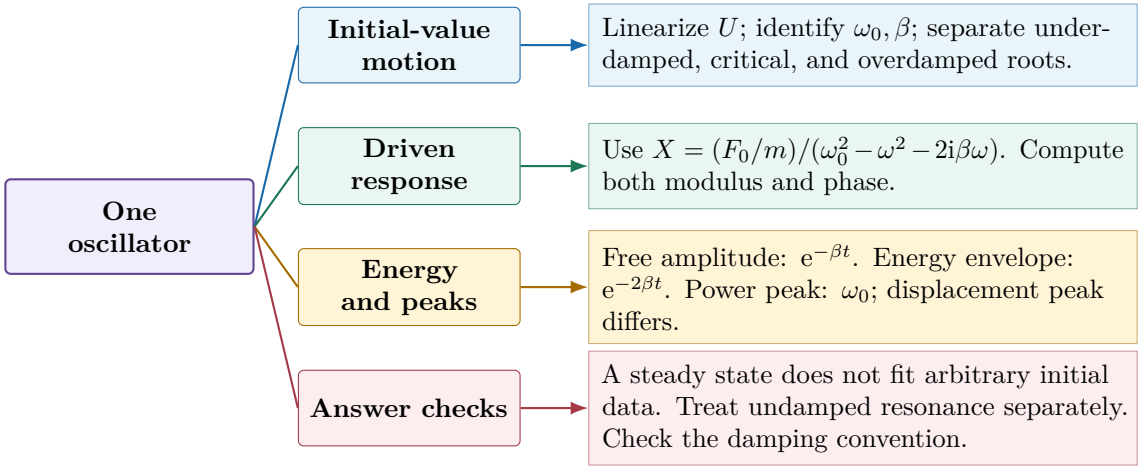


Figure 2.6: Chapter 2 mind map: choose the measured quantity before identifying a resonance.

3 Coupled Oscillators and Normal Coordinates

Source map: Syllabus Unit 2; connections to generalized eigenvalue problems in Algebra 2M.

3.1 Two masses and three longitudinal springs

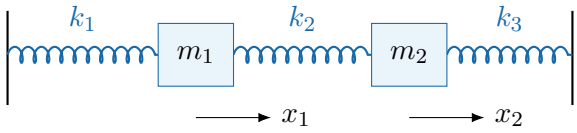


Figure 3.1: Displacements are measured from the static equilibrium configuration. The middle spring extension is $x_2 - x_1$.

The kinetic and quadratic potential energies are

$$T = \tfrac{1}{2}m_1\dot{x}_1^2 + \tfrac{1}{2}m_2\dot{x}_2^2, \qquad U = \tfrac{1}{2}k_1x_1^2 + \tfrac{1}{2}k_2(x_2 - x_1)^2 + \tfrac{1}{2}k_3x_2^2.$$

Taking $-\partial U/\partial x_j$ yields

$$M\ddot{\boldsymbol{x}} + K\boldsymbol{x} = 0, \quad M = \begin{pmatrix} m_1 & 0 \\ 0 & m_2 \end{pmatrix}, \quad K = \begin{pmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 + k_3 \end{pmatrix}.$$

The off-diagonal entries are negative because moving one mass right pulls its neighbor right rather than restoring that neighbor to the origin. A trial $\boldsymbol{x} = \boldsymbol{a}\mathrm{e}^{-\mathrm{i}\omega t}$ gives $\det(K - \omega^2 M) = 0$.

Key Result: The general two-frequency formula

With $A = (k_1 + k_2)/m_1$ and $B = (k_2 + k_3)/m_2$,

$$\omega_{\pm}^2 = \frac{A + B \pm \sqrt{(A - B)^2 + 4k_2^2/(m_1m_2)}}{2}.$$

The amplitude ratio follows from either row:

$$\frac{a_2}{a_1} = \frac{k_1 + k_2 - m_1\omega^2}{k_2} \quad (k_2 \neq 0).$$

If $k_2 = 0$, solve two independent oscillators; do not divide by zero. A zero eigenvalue describes an unrestrained coordinate, not an oscillation with finite period.

3.2 Symmetric modes and modal initial data

For $m_1 = m_2 = m$, $k_1 = k_3 = k$, and $k_2 = \kappa$,

$$\boldsymbol{a}_s = (1, 1)^T, \quad \omega_s^2 = k/m; \qquad \boldsymbol{a}_a = (1, -1)^T, \quad \omega_a^2 = (k + 2\kappa)/m.$$

In the symmetric mode, the middle spring is unextended. In the antisymmetric mode it changes length by twice the displacement, raising the frequency. Define $q_s = (x_1 + x_2)/2$ and $q_a = (x_1 - x_2)/2$. Adding and subtracting Newton's equations gives independent equations for q_s and q_a. These coordinates need not be mass-normalized to decouple; normalization matters when computing modal forces and energy.

Worked Example: A motion that is not one normal mode

Release $(x_1, x_2) = (A, 0)$ from rest. Then $q_s(0) = q_a(0) = A/2$ and

$$x_1 = \tfrac{A}{2}(\cos\omega_s t + \cos\omega_a t) = A\cos(\bar{\omega}t)\cos(\Delta\omega t/2),$$
$$x_2 = \tfrac{A}{2}(\cos\omega_s t - \cos\omega_a t) = A\sin(\bar{\omega}t)\sin(\Delta\omega t/2),$$

where $\bar{\omega} = (\omega_a + \omega_s)/2$ and $\Delta\omega = \omega_a - \omega_s > 0$. Both coordinates obey the original initial data. The slow exchange timescale is $\pi/\Delta\omega$ for the first maximum of the second displacement envelope. The intensity/energy-envelope repetition time is $2\pi/\Delta\omega$.

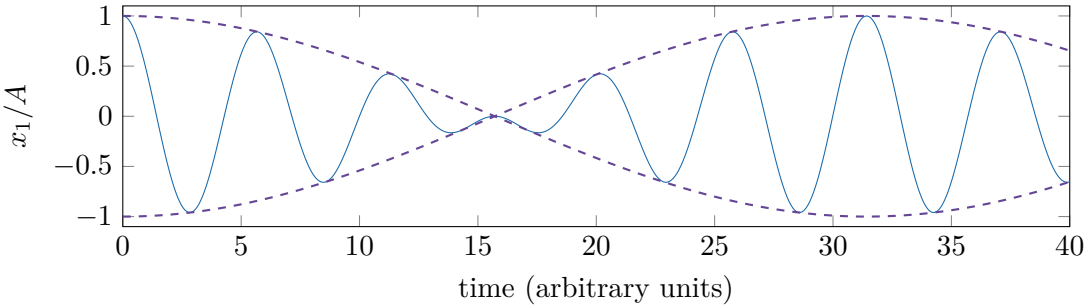


Figure 3.2: Two nearby mode frequencies produce a fast carrier and a slow envelope.

For weak coupling, $\omega_a \simeq \omega_s(1 + \kappa/k)$ and $\Delta\omega \simeq \kappa/(m\omega_s)$. A clean cycle-averaged transfer of local oscillator energy is then approximately described by $\cos^2(\Delta\omega t/2)$ and $\sin^2(\Delta\omega t/2)$. The coupling spring also stores energy. Exact total energy is conserved, but assigning all coupling energy to one mass and claiming exact independent energy fractions is generally wrong.

Worked Example: Unequal masses require a mass inner product

Let $M = \mathrm{diag}(m, 2m)$ and $K = k\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$. With $s = m\omega^2/k$, the determinant is $(2-s)(2-2s) - 1 = 2s^2 - 6s + 3$. Thus $s_\pm = (3 \pm \sqrt{3})/2$. Choose $\boldsymbol{a}_\pm = (1, 2 - s_\pm)^T$. Their ordinary dot product is $1/2$, not zero, but

$$\boldsymbol{a}_+^T M \boldsymbol{a}_- = m\{1 + 2(2 - s_+)(2 - s_-)\} = 0.$$

To project initial data, divide each vector by $\sqrt{m[1 + 2(2 - s_\pm)^2]}$ first. This is the same self-adjoint structure used in linear algebra with a weighted inner product.

3.3 Transverse coupling, pre-tension, and the slinky limit

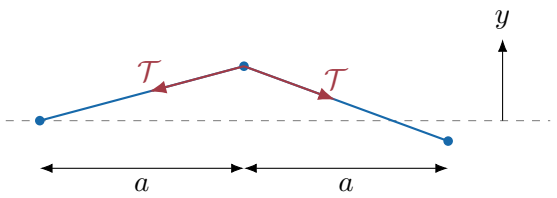


Figure 3.3: The vertical force is the difference of the vertical components of the tensions on neighboring segments.

Let equilibrium spacing be a and vertical displacements y_j. If the segment slopes satisfy $|y_{j+1} - y_j|/a \ll 1$ and the equilibrium tension is $\mathcal{T}$, then $\sin\theta \simeq \tan\theta = (y_{j+1} - y_j)/a$. Newton's law gives

$$m\ddot{y}_j = \frac{\mathcal{T}}{a}(y_{j+1} - 2y_j + y_{j-1}).$$

The effective transverse coupling is $\mathcal{T}/a$, not automatically the longitudinal spring stiffness. For a spring of stiffness K and natural length ℓ_0 stretched to a, $\mathcal{T} = K(a - \ell_0)$, so $K_\perp = K(1 - \ell_0/a)$. The change of tension due to a small transverse slope is second order and can be neglected in the linear force.

Key Result: Why the slinky limit needs a condition

If the equilibrium spring length is much greater than its natural length ($\ell_0/a \ll 1$), then $\mathcal{T} \simeq Ka$ and $K_\perp \simeq K$. Longitudinal and transverse equations then have the same coupling scale. Without pre-tension ($a = \ell_0$), the linear transverse restoring coefficient vanishes; the leading force is nonlinear. A taut-string equation is not justified for an unstretched chain simply by making the displacement small.

Worked Example: Two transverse masses

Three equal pre-tensioned segments of length a connect two equal masses to fixed endpoints. The equations are the equal-three-spring equations with $k = \kappa = \mathcal{T}/a$. Thus $\omega_s = \sqrt{\mathcal{T}/(ma)}$ and $\omega_a = \sqrt{3\mathcal{T}/(ma)}$. The mode shapes are $(1, 1)$ and $(1, -1)$, but the physical displacements are vertical. If the tension doubles at fixed a, both frequencies scale by $\sqrt{2}$.

3.4 Driving selects modes, not just frequencies

For mass-normalized modes, a force $\boldsymbol{f}(t)$ produces

$$\ddot{Q}_r + \omega_r^2 Q_r = \boldsymbol{a}_r^T \boldsymbol{f}(t).$$

A symmetric force (F, F) has zero projection on an antisymmetric mode. A resonance denominator alone does not guarantee a response: the corresponding forcing numerator can vanish. If damping is proportional to M, each mode acquires the same $2\beta\dot{Q}_r$; a general damping matrix need not diagonalize in the conservative mode basis.

Standard Workflow: Coupled-system questions

Draw the equilibrium geometry; express every spring extension; construct M and K; solve the generalized eigenvalue equation; calculate and normalize mode shapes; project the data or force; reconstruct the coordinates; check uncoupled, symmetric, and zero-frequency limits.

3.5 Derivation Workshop: Decoupling, Selection, and Antiresonance

3.5.1 Why mass-normalized coordinates truly decouple the dynamics

Let A contain mass-normalized eigenvectors. Substitute $\boldsymbol{x} = A\boldsymbol{Q}$ into $M\ddot{\boldsymbol{x}} + K\boldsymbol{x} = \boldsymbol{f}$ and premultiply by A^T:

$$\ddot{\boldsymbol{Q}} + \operatorname{diag}(\omega_1^2, \ldots, \omega_N^2)\boldsymbol{Q} = A^T \boldsymbol{f}.$$

This uses both $A^T M A = I$ and $A^T K A = \operatorname{diag}(\omega_r^2)$. The energy becomes

$$E = \tfrac{1}{2} \sum_r (\dot{Q}_r^2 + \omega_r^2 Q_r^2).$$

There are no cross terms in this basis. The force projection is $\boldsymbol{a}_r^T \boldsymbol{f}$, while a displacement projection is $\boldsymbol{a}_r^T M \boldsymbol{x}$; mixing these two projections inserts an incorrect extra mass factor.

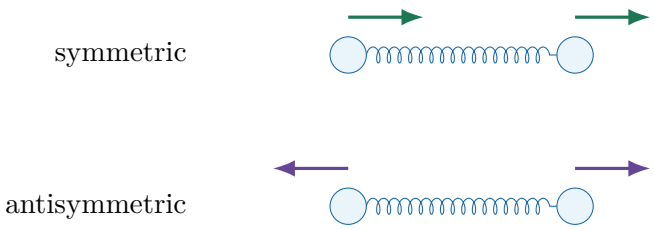


Figure 3.4: For a symmetric pair, equal displacements leave the coupling spring unchanged, while opposite displacements stretch it. Arrow lengths show relative amplitudes, not absolute scale.

3.5.2 A geometric expansion for transverse coupling

For one spring with endpoint separation $(a, \Delta y)$, its length is

$$\ell = \sqrt{a^2 + (\Delta y)^2} = a + \frac{(\Delta y)^2}{2a} + O((\Delta y)^4/a^3).$$

Its vertical force is $K(\ell - \ell_0)\Delta y/\ell$. Expanding to first order gives $K(a - \ell_0)\Delta y/a$, which is $\mathcal{T}\Delta y/a$. If $a = \ell_0$, the first nonzero term is $K(\Delta y)^3/(2a^2)$. This derivation explains precisely why pre-tension is required for a nonzero linear transverse frequency. A linearized model with zero transverse stiffness has lost the leading nonlinear term; it does not imply that a finite sideways displacement feels no force at all.

3.5.3 Antiresonance: one coordinate can stop while another moves

For a symmetric pair driven only at mass 1, neglect damping and assume the drive is not at an eigenfrequency. Put $d = k + \kappa - m\omega^2$. The amplitude equations are

$$\begin{pmatrix} d & -\kappa \\ -\kappa & d \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} = \begin{pmatrix} F_0 \\ 0 \end{pmatrix}, \quad X_1 = \frac{F_0 d}{d^2 - \kappa^2}, \quad X_2 = \frac{F_0 \kappa}{d^2 - \kappa^2}.$$

At $\omega^2 = (k+\kappa)/m$, $d = 0$, but $d^2 - \kappa^2 = -\kappa^2 \neq 0$ for nonzero coupling. Hence $X_1 = 0$ and $X_2 = -F_0/\kappa$. The driven mass is stationary because the coupling force from mass 2 balances the applied force. This is a zero of a response numerator, not a zero eigenfrequency. It occurs between the two resonance frequencies.

Worked Example: An exact antiresonant motion

Take $m = 1\,\mathrm{kg}$, $k = 4\,\mathrm{N\,m^{-1}}$, $\kappa = 6\,\mathrm{N\,m^{-1}}$, and $F_0 = 0.60\,\mathrm{N}$. The free frequencies are 2 and $4\,\mathrm{s^{-1}}$. At the intermediate drive $\omega = \sqrt{10}\,\mathrm{s^{-1}}$, $X_1 = 0$ and $X_2 = -0.10\,\mathrm{m}$. In row 1, $-\kappa X_2 = 0.60\,\mathrm{N}$ balances the applied force; row 2 is $dX_2 = 0$. The algebra checks both equations. With damping, the cancellation is generally incomplete and the response minimum need not be exactly zero.

3.5.4 Beats: what is exact and what is averaged

The trigonometric identities for x_1, x_2 are exact for the specified equal-mode initial amplitudes. Approximate local-energy exchange requires an additional scale separation $\Delta\omega \ll \bar{\omega}$. Over a few fast cycles but much less than one beat period, the envelopes are nearly constant, allowing the local kinetic and outer-spring energies to follow approximately $\cos^2(\Delta\omega t/2)$ and $\sin^2(\Delta\omega t/2)$. The exact conserved energy also contains $\kappa(x_2 - x_1)^2/2$ and derivative terms involving the changing envelope. This is why a maximum of a displacement envelope should not be called an exact complete transfer of all system energy without qualification.

Standard Workflow: A standard coupled-system solution

1. Draw equilibrium positions and assign a consistent positive displacement direction.
2. Write every spring extension; calculate M and K from T and U or Newton's law.
3. Solve $\det(K-\omega^2 M)=0$, then find each eigenvector from a row equation.
4. Normalize using M; project both initial data or the external force.
5. Reconstruct coordinates. If a response vanishes, inspect its numerator before claiming a missing mode.
6. Check the uncoupled, symmetric, and translational limits and the total energy.

3.6 Chapter Mind Map

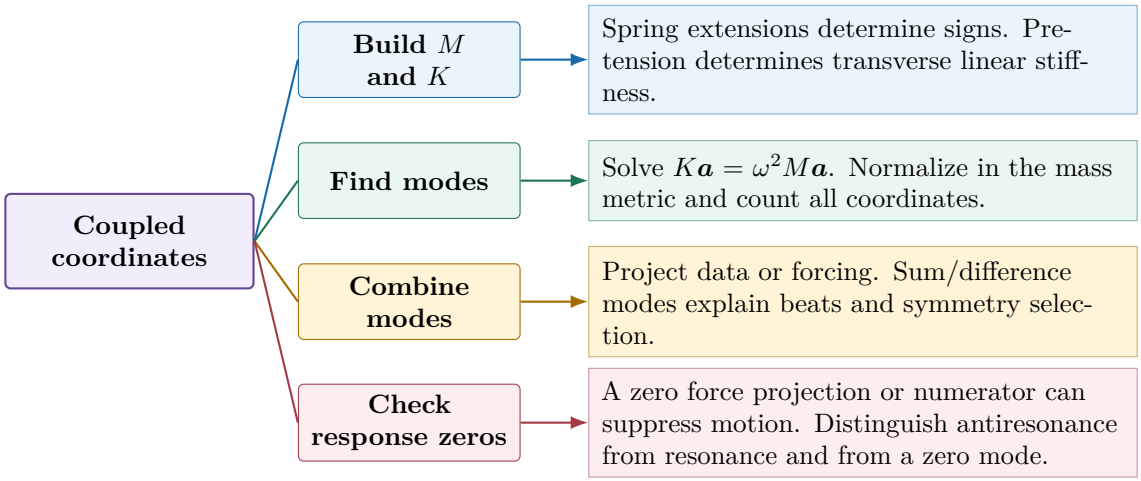


Figure 3.5: Chapter 3 mind map: the mode basis organizes both free and driven motion.

4 Many Degrees of Freedom, Dispersion, and Phonons

Source map: Syllabus Unit 3; self-contained discrete-chain derivations.

4.1 Monatomic chain: derive the dispersion before interpreting it

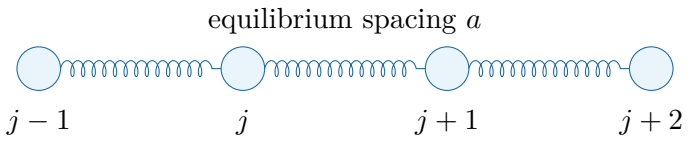


Figure 4.1: A nearest-neighbor longitudinal chain with mass m, spring stiffness K, and spacing a.

For displacement u_j from equilibrium,

$$m\ddot{u}_j = K(u_{j+1}-2u_j+u_{j-1}).$$

Set $u_j = U\mathrm{e}^{\mathrm{i}(kja-\omega t)}$. The neighboring factors are $\mathrm{e}^{\pm \mathrm{i}ka}$, giving

$$-m\omega^2 = K(2\cos ka-2) = -4K\sin^2(ka/2).$$

Key Result: Monatomic lattice dispersion

$$\omega(k) = 2\sqrt{K/m}\,|\sin(ka/2)|, \qquad -\pi/a \le k \le \pi/a.$$

Wavevectors differing by $2\pi/a$ give the same displacement at every site. The first Brillouin zone removes this redundancy; it is not a restriction that the wavelength of an arbitrary continuous function must satisfy.

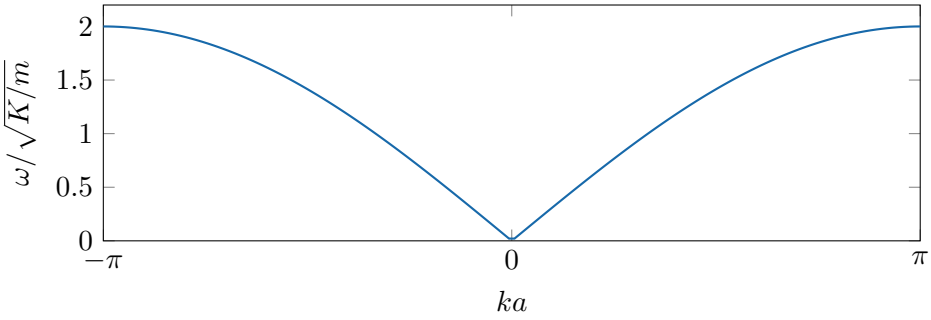


Figure 4.2: The lattice has a finite frequency band and a nonlinear dispersion relation away from long wavelengths.

For $0 < ka < \pi$,

$$v_p = \frac{\omega}{k}, \qquad v_g = \frac{\mathrm{d}\omega}{\mathrm{d}k} = a\sqrt{K/m}\cos(ka/2).$$

At $ka \ll 1$, $\omega \simeq c_s|k|$ with $c_s = a\sqrt{K/m}$. At the zone edge, $v_g = 0$ although $\omega \neq 0$: the mode oscillates, but a narrow packet centered there has zero first-order envelope velocity.

4.2 Boundary conditions count the allowed modes

Definition: Periodic and fixed-end chains

For N masses in a ring, $u_{j+N} = u_j$ implies $\mathrm{e}^{\mathrm{i}kNa} = 1$ and $k_s = 2\pi s/(Na)$, with N inequivalent values modulo $2\pi/a$. The $k = 0$ mode is uniform translation and has zero restoring force.
For N moving masses between fixed sites $u_0 = u_{N+1} = 0$, use

$$u_j^{(n)} \propto \sin\frac{n\pi j}{N+1}, \quad k_n = \frac{n\pi}{(N+1)a}, \quad n = 1, \dots, N.$$

There is no translational zero mode because the supports break translation symmetry.

For the fixed chain, discrete sine orthogonality gives $\sum_{j=1}^{N} \sin(k_n ja)\sin(k_m ja) = (N+1)\delta_{nm}/2$. This proves completeness of the N orthogonal eigenvectors, rather than merely guessing some modes.

Worked Example: A five-mass chain

Take $N = 5$, $K/m = 100\,\mathrm{s}^{-2}$. Then $\omega_n = 20\sin(n\pi/12)\,\mathrm{s}^{-1}$, giving approximately 5.176, 10.000, 14.142, 17.321, and 19.319. Mode $n = 3$ has site pattern $(1, 0, -1, 0, 1)$ up to a scale. The number of independent modes equals the number of moving coordinates, five, not the number of springs, six.

4.3 Continuum limit and sound

Replace $u_j(t)$ by $u(x,t)$ with $x = ja$. Taylor-expand:

$$u(x+a) - 2u(x) + u(x-a) = a^2 u_{xx} + \frac{a^4}{12}u_{xxxx} + \cdots.$$

Keeping only the first term gives $u_{tt} = c_s^2 u_{xx}$. This requires spatial variation over lengths much greater than a. The higher derivatives produce lattice dispersion and cannot be ignored for wavelengths of a few lattice spacings. For a transverse chain, substitute $K = \mathcal{T}/a$ and $m = \mu a$ to get $c_s^2 = \mathcal{T}/\mu$.

Worked Example: A dispersive packet

With $a = 0.010\,\mathrm{m}$, $K/m = 10^4\,\mathrm{s}^{-2}$, and carrier $ka = 2\pi/3$, $\omega = 100\sqrt{3}\,\mathrm{s}^{-1}$. Thus $v_p = 3\sqrt{3}/(2\pi) = 0.827\,\mathrm{m\,s}^{-1}$, whereas $v_g = 0.500\,\mathrm{m\,s}^{-1}$. The long-wave speed is $1.00\,\mathrm{m\,s}^{-1}$. A wave crest and a packet envelope do not generally move together.

4.4 Binary chain: define the cell before writing the phase

Let adjacent-site spacing be a and unit-cell length be $2a$. In cell ℓ, mass M at $2\ell a$ has displacement u_ℓ, and mass m at $(2\ell+1)a$ has displacement v_ℓ. Neighboring springs all have stiffness K:

$$M\ddot{u}_\ell = K(v_\ell + v_{\ell-1} - 2u_\ell),$$
$$m\ddot{v}_\ell = K(u_\ell + u_{\ell+1} - 2v_\ell).$$

Take $u_\ell = U\mathrm{e}^{\mathrm{i}(2\ell aq-\omega t)}$ and $v_\ell = V\mathrm{e}^{\mathrm{i}(2\ell aq-\omega t)}$. Both amplitudes use the cell origin; the position offset of the second atom is absorbed in V. The matrix is

$$\begin{pmatrix} 2K - M\omega^2 & -K(1+\mathrm{e}^{-2\mathrm{i}qa}) \\ -K(1+\mathrm{e}^{2\mathrm{i}qa}) & 2K - m\omega^2 \end{pmatrix} \begin{pmatrix} U \\ V \end{pmatrix} = 0.$$

Its determinant gives $Mm\omega^4 - 2K(M+m)\omega^2 + 4K^2\sin^2(qa) = 0$.

Key Result: Acoustic and optical branches

$$\omega_\pm^2 = K\left(\frac{1}{M}+\frac{1}{m}\right) \pm K\sqrt{\left(\frac{1}{M}+\frac{1}{m}\right)^2 - \frac{4}{Mm}\sin^2(qa)}, \quad |q| \le \frac{\pi}{2a}.$$

The minus sign is acoustic, with $\omega_-(0) = 0$ and $U = V$ at $q = 0$. The plus sign is optical, with $\omega_+^2(0) = 2K(1/M + 1/m)$ and $MU + mV = 0$ there. Optical means finite frequency at long wavelength, not necessarily visible light or guaranteed optical activity.

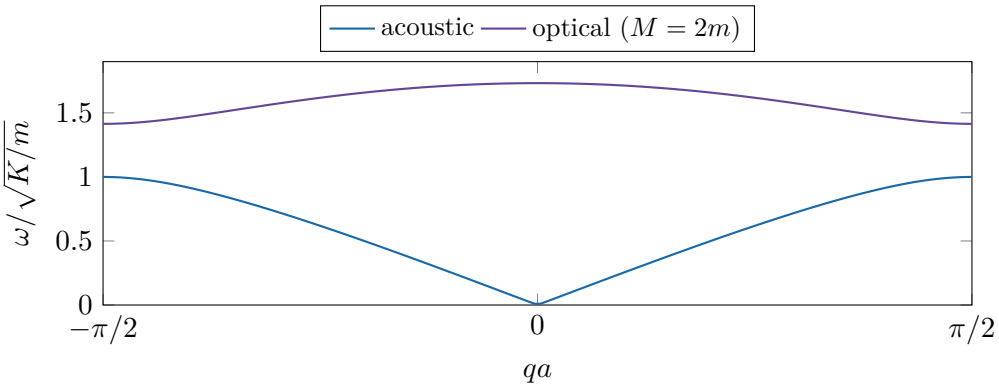


Figure 4.3: Two masses per cell produce two frequency branches. The gap is nonzero when the masses differ.

If $M > m$, the zone-edge limits are $\omega_-^2 = 2K/M$ and $\omega_+^2 = 2K/m$. No extended normal mode has frequency in the interval between these band edges. When $M = m$, the gap closes: this is the monatomic dispersion folded into a smaller Brillouin zone, not a new physical gap. At small q, expand the square root to find

$$\omega_- \simeq a\sqrt{\frac{2K}{M+m}}\,|q|.$$

Worked Example: Check the binary-chain limits

For $M = 3m$, the acoustic zone-edge frequency is $\sqrt{2K/(3m)}$, the optical edge is $\sqrt{2K/m}$, and the optical zone-center frequency is $\sqrt{8K/(3m)}$. At the optical center, $V = -3U$: the lighter mass moves more, keeping the cell's center of mass fixed. There are $2N_c$ modes for N_c cells, N_c on each branch.

4.5 Forced many-coordinate motion and mode selection

If $M\ddot{\boldsymbol{u}} + 2\beta M\dot{\boldsymbol{u}} + K\boldsymbol{u} = \mathrm{Re}(\boldsymbol{f}\mathrm{e}^{-\mathrm{i}\omega t})$, mass-normalized modes yield

$$\widetilde{Q}_r = \frac{\boldsymbol{a}_r^T \boldsymbol{f}}{\omega_r^2 - \omega^2 - 2\mathrm{i}\beta\omega}, \qquad \widetilde{\boldsymbol{u}} = \sum_r \boldsymbol{a}_r \widetilde{Q}_r.$$

A local drive excites every mode with nonzero displacement at the drive site. A spatially periodic drive selects matching wavevectors in a translationally invariant lattice. At an undamped modal resonance, use a growing resonant solution if the force projection is nonzero.

Supplementary Background: What is a phonon?

A normal mode is a classical collective coordinate. Upon quantization its energy levels are $E_r = \hbar\omega_r(n_r + 1/2)$, and one added quantum is a phonon. The lattice dispersion specifies phonon frequencies and group velocities. Quantization is a connection mentioned by the syllabus, not a change to the classical equations derived above.

Standard Workflow: Lattice problems

Specify site spacing and cell length; derive nearest-neighbor equations; insert a phase-consistent trial wave; solve the determinant; count modes using boundaries; distinguish branch from wavevector; differentiate the correct branch for v_g; check $q \to 0$, zone edge, and equal-mass limits.

4.6 Derivation Workshop: Mode Counting and Band Interpretation

4.6.1 Periodic orthogonality and reciprocal-vector equivalence

For N sites, periodic modes have $k_s = 2\pi s/(Na)$ and site values $\mathrm{e}^{\mathrm{i}k_s ja}$. Their inner product contains

$$S_{sr} = \sum_{j=0}^{N-1} \mathrm{e}^{2\pi\mathrm{i}(r-s)j/N}.$$

If $r = s$ modulo N, every term is one and $S_{sr} = N$. Otherwise this is a geometric sum with ratio $z \neq 1$ but $z^N = 1$, hence $S_{sr} = (1 - z^N)/(1 - z) = 0$. Thus the discrete Fourier basis is orthogonal and complete. Wavevectors shifted by $G = 2\pi/a$ have exactly the same site phases, not merely the same frequency. Choosing one Brillouin zone prevents overcounting the same physical displacement.

For a real displacement, the complex coefficients obey $U(-k) = U(k)^*$. Real sine and cosine patterns are alternative coordinates for each conjugate pair. A self-conjugate wavevector such as $k = 0$ or the zone edge for an even-site ring is counted once, not as two independent traveling patterns.

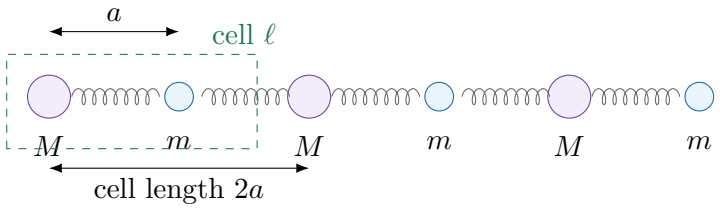


Figure 4.4: Adjacent spacing a and unit-cell length $2a$ must be distinguished before deriving a binary-chain phase convention.

4.6.2 Binary-chain acoustic slope, without skipping the expansion

Let $S = 1/M + 1/m = (M+m)/(Mm)$. The acoustic branch is

$$\omega_-^2 = KS - KS\sqrt{1 - \frac{4\sin^2 qa}{MmS^2}}.$$

For $|qa| \ll 1$, use $\sin^2 qa \simeq q^2a^2$ and $\sqrt{1-\varepsilon} \simeq 1 - \varepsilon/2$. Then

$$\omega_-^2 \simeq KS\frac{2q^2a^2}{MmS^2} = \frac{2Ka^2}{M+m}q^2, \qquad c_s = a\sqrt{\frac{2K}{M+m}}.$$

The equivalent cell mass density is $(M+m)/(2a)$, while the long-wavelength elastic scale produces the same result. Setting $M = m$ recovers $a\sqrt{K/m}$, a necessary consistency check.

4.6.3 Band-edge eigenvectors: avoid an indeterminate amplitude ratio

At $q = \pi/(2a)$, $1 + \mathrm{e}^{\pm 2iqa} = 0$, so the binary dynamic matrix is diagonal. If $M > m$, the acoustic edge has $\omega^2 = 2K/M$ and $V = 0$: the heavy sublattice moves while the light sublattice has a node. The optical edge has $\omega^2 = 2K/m$ and $U = 0$. A generic formula for V/U can become $0/0$ or divide by zero here; return to the matrix equations.

Worked Example: Recognize a branch from measured site motion

An alternating chain with $M = 4m$ is observed at a zone-edge frequency. Heavy sites move while light sites remain still. The branch is acoustic and $\omega = \sqrt{2K/M} = \sqrt{K/(2m)}$. At the optical edge, the heavy sites would instead be still and $\omega = \sqrt{2K/m}$, twice as large. Frequency and eigenvector information reinforce each other; either alone can be insufficient if the spring constant is unknown.

4.6.4 A zero mode changes the forced-response question

The uniform mode of an unconstrained periodic chain has no restoring force. Adding the Newton equations cancels all internal spring forces:

$$Nm\ddot{R} = \sum_j f_j(t), \qquad R = \frac{1}{N}\sum_j u_j.$$

A constant nonzero net force accelerates the center of mass, so no bounded static equilibrium exists. If a static force pattern has zero sum, it is orthogonal to the zero mode and a static distortion can exist, but it is determined only up to a uniform translation until a reference position is fixed. A harmonic drive at nonzero ω can still produce a finite zero-mode displacement proportional to $-1/\omega^2$.

Standard Workflow: Dispersion and lattice questions

1. State the site spacing, cell length, number of cells, and boundary conditions.
2. Derive the nearest-neighbor equations; define where each complex amplitude's phase is referenced.
3. Solve the dynamic determinant for ω^2, not for an assumed linear $\omega(k)$.
4. Count the allowed wavevectors and branches, including zero modes and conjugate pairs.
5. Calculate eigenvectors and inspect exceptional points directly in the matrix.
6. Derive v_g from a branch derivative; inspect the long-wave, zone-edge, and equal-mass limits.

4.7 Chapter Mind Map

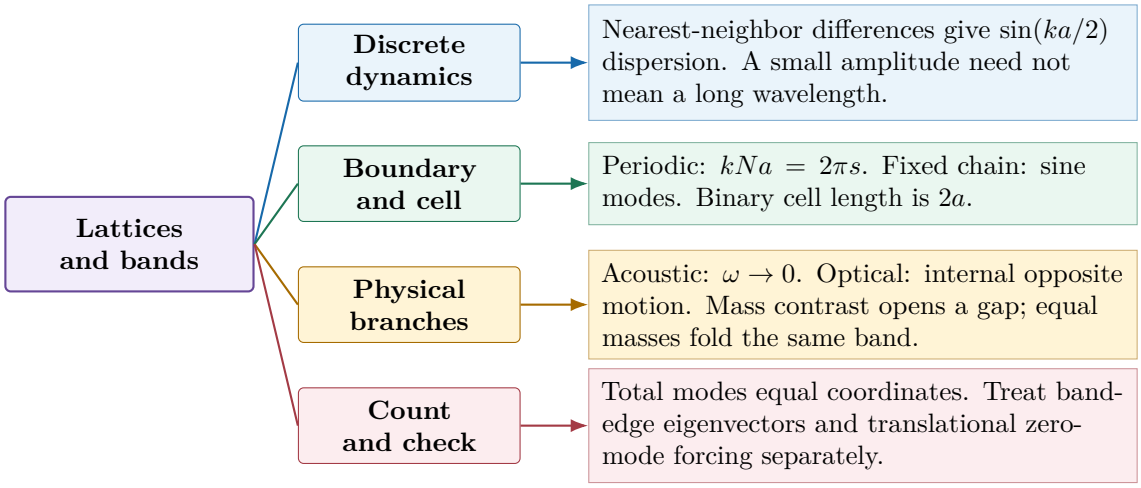


Figure 4.5: Chapter 4 mind map: geometry, boundaries, and cell choice determine how dispersion is interpreted.

5 String Waves, Boundaries, and Fourier Synthesis

Source map: Syllabus Unit 4; Fourier conventions linked to Peatross and Ware Section 0.4.

5.1 The string equation is a force balance

Let $y(x,t)$ be the small transverse displacement of a taut string, with tension $\mathcal{T}$ and mass per length μ. For an element $\mathrm{d}x$, the difference of vertical tension components is $\mathcal{T}[y_x(x+\mathrm{d}x) - y_x(x)]$. Its mass is $\mu \mathrm{d}x$, hence

$$\mu y_{tt} = \mathcal{T} y_{xx}, \qquad v = \sqrt{\mathcal{T}/\mu}.$$

We assumed $|y_x| \ll 1$, constant tension, negligible bending stiffness, and negligible damping. The disturbance is transverse, although the net energy travels along the string. A string particle does not move along the entire string with the wave speed.

5.2 Separation and endpoint conditions

Set $y = X(x)Q(t)$. Dividing $X\ddot{Q} = v^2 X'' Q$ by $v^2 XQ$ yields a constant $-k^2$:

$$X'' + k^2 X = 0, \qquad \ddot{Q} + \omega^2 Q = 0, \qquad \omega = vk.$$

The boundary conditions determine k; the differential equation alone does not quantize frequencies.

Key Result: Three common endpoint combinations

Ends	Spatial function	Allowed k
fixed–fixed	$\sin(kx)$	$n\pi/L$, $n = 1, 2, \ldots$
fixed–free	$\sin(kx)$	$(n + 1/2)\pi/L$, $n = 0, 1, \ldots$
free–free	$\cos(kx)$	$n\pi/L$, $n = 0, 1, \ldots$

A fixed end has $y = 0$; an ideal free end has $y_x = 0$, since the transverse force is zero. The free–free $n = 0$ mode is rigid transverse translation: its time dependence is $A + Bt$, not a nonzero-frequency sine.

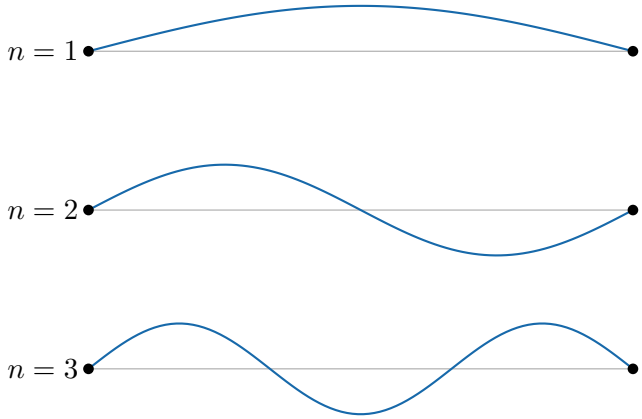


Figure 5.1: The first three fixed–fixed string shapes, each with zero endpoint displacement.

For fixed–fixed boundaries, $\lambda_n = 2L/n$ and $f_n = nv/(2L)$. There are $n-1$ internal nodes. At a node the displacement is always zero; at an antinode the amplitude is maximal. A point that happens to cross $y = 0$ at one instant is not necessarily a node.

Worked Example: Infer tension from a mode pattern

A 1.20 m string of density $0.0050\,\mathrm{kg\,m^{-1}}$ has four loops at 200 Hz. Four loops means $n = 4$, so $\lambda = 2L/4 = 0.600\,\mathrm{m}$, $v = f\lambda = 120\,\mathrm{m\,s^{-1}}$, and $\mathcal{T} = \mu v^2 = 72.0\,\mathrm{N}$. Its fundamental is 50.0 Hz. Counting internal nodes as the mode number would give the wrong tension.

5.3 Arbitrary displacement and velocity data

For initial shape $f(x)$ and velocity $g(x)$ on a fixed string,

$$y(x,t) = \sum_{n=1}^{\infty} \left[A_n \cos \omega_n t + B_n \sin \omega_n t\right] \sin \frac{n\pi x}{L}.$$

Multiply the initial conditions by the mth sine and integrate. Orthogonality isolates one coefficient:

Key Result: Fourier coefficients for initial data

$$A_n = \frac{2}{L}\int_0^L f(x) \sin \frac{n\pi x}{L}\mathrm{d}x, \qquad B_n = \frac{2}{L\omega_n}\int_0^L g(x) \sin \frac{n\pi x}{L}\mathrm{d}x.$$

Displacement and velocity require separate projections. For mixed boundaries, use the corresponding half-integer sine basis; do not keep the fixed–fixed frequencies.

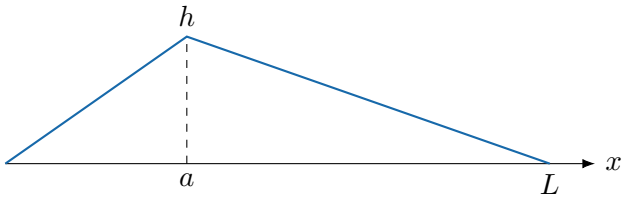


Figure 5.2: A triangular pluck at $x = a$. Its lack of smoothness populates many normal modes.

Worked Example: Derive the complete triangular-pluck spectrum

Let $f(x) = hx/a$ for $0 < x < a$ and $f(x) = h(L-x)/(L-a)$ for $a < x < L$, with zero initial velocity. Splitting the integral at a gives

$$A_n = \frac{2h}{L}\left[\frac{1}{a}\int_0^a x \sin k_n x\mathrm{d}x + \frac{1}{L-a}\int_a^L (L-x)\sin k_n x\mathrm{d}x\right]$$

$$= \frac{2hL^2}{\pi^2 n^2 a(L-a)} \sin\frac{n\pi a}{L}, \qquad B_n = 0.$$

The antiderivative $\int x \sin kx \mathrm{d}x = -x\cos kx/k + \sin kx/k^2$ explains the n^{-2} decay; the cosine endpoint terms cancel between segments. For a midpoint pluck, $A_n = 8h\sin(n\pi/2)/(\pi^2 n^2)$, so all even modes vanish. For a pluck at $L/3$, every third mode vanishes. The selection rule comes from the sine factor, not from a new dispersion relation.

Worked Example: Initial velocity excites a different quadrature

Let $y(x,0) = 0$ and $y_t(x,0) = V\sin(3\pi x/L)$. Then $A_n = 0$ and $B_n = (V/\omega_3)\delta_{n3}$. Therefore

$$y(x,t) = \frac{V}{\omega_3}\sin\omega_3 t \sin\frac{3\pi x}{L}.$$

At $t = 0$ the shape is flat even though the string is not at rest. Inferring zero energy from an instantaneous flat shape ignores kinetic energy.

5.4 Mode energy and Parseval's physical meaning

The instantaneous string energy is

$$E = \frac{1}{2}\int_0^L [\mu y_t^2 + \mathcal{T} y_x^2]\mathrm{d}x.$$

For a single fixed–fixed mode $y = q_n(t)\sin k_n x$, sine and cosine integrals both equal $L/2$, giving

$$E_n = \frac{\mu L}{4}(\dot{q}_n^2 + \omega_n^2 q_n^2) = \frac{\mu L}{4}\omega_n^2(A_n^2 + B_n^2).$$

Cross terms vanish between different modes. Consequently $E = \sum_n E_n$ for sufficiently regular finite-energy data. This is Parseval orthogonality expressed as physical energy accounting.

Worked Example: Check a pluck by two independent energies

For the triangular pluck, the slopes are h/a and $-h/(L-a)$. Thus

$$E(0) = \frac{\mathcal{T}}{2}\left(\frac{h^2}{a} + \frac{h^2}{L-a}\right) = \frac{\mathcal{T}h^2 L}{2a(L-a)}.$$

The modal sum must reproduce this value. For $a = L/2$, $E = 2\mathcal{T}h^2/L$. Missing the $1/2$ in the potential-energy density doubles the answer.

5.5 Driven strings and resonance selection

For a transverse force density $f(x)\cos\omega t$ and viscous damping $2\beta\mu y_t$,

$$\mu y_{tt} + 2\beta\mu y_t - \mathcal{T} y_{xx} = f(x)\cos\omega t.$$

The modal force coefficient is $f_n = (2/L)\int_0^L f(x)\sin(k_n x)\mathrm{d}x$, and the steady modal amplitude is

$$\widetilde{q}_n = \frac{f_n/\mu}{\omega_n^2 - \omega^2 - 2\mathrm{i}\beta\omega}.$$

A point drive $F\delta(x - x_d)$ has $f_n = 2F\sin(k_n x_d)/L$. It cannot excite a mode whose node coincides with x_d.

Error Check: What Fourier convergence does and does not say

At an interior jump, the Fourier series converges to the average of the one-sided limits under the usual piecewise-smooth conditions. High-mode truncations can show Gibbs overshoot near a jump. Incompatible nonzero endpoint displacement cannot satisfy a permanently fixed boundary at that endpoint; state the compatibility requirement instead of silently forcing the data to fit.

Standard Workflow: Standing-wave initial-value problems

Derive v, impose endpoint conditions, identify the orthogonal mode basis, project both initial shape and initial velocity, evolve each coefficient, and reconstruct. Verify the initial data and use energy or a symmetry selection rule as an independent check.

5.6 Derivation Workshop: Boundaries, Projections, and Energy

5.6.1 Why the boundary conditions produce orthogonal functions

Let $X_n'' = -k_n^2 X_n$ and $X_m'' = -k_m^2 X_m$. Multiply by X_m and X_n respectively, subtract, and integrate:

$$(k_m^2 - k_n^2)\int_0^L X_m X_n \mathrm{d}x = [X_m X_n' - X_n X_m']_0^L.$$

At a fixed endpoint all admissible X vanish; at a free endpoint all admissible X' vanish. For the same fixed/free conditions imposed on both modes, the boundary term is zero. Distinct k_n^2 then imply orthogonality. It is the boundary term, not just the differential equation, that makes the problem self-adjoint and gives an orthogonal expansion.

5.6.2 The triangular-pluck integral in full

Use $k_n = n\pi/L$, $f = hx/a$ on the left, and $f = h(L-x)/(L-a)$ on the right. The two elementary integrals are

$$I_1 = \int_0^a x\sin(k_n x)\mathrm{d}x = -\frac{a\cos(k_n a)}{k_n} + \frac{\sin(k_n a)}{k_n^2},$$
$$I_2 = \int_a^L (L-x)\sin(k_n x)\mathrm{d}x = \frac{(L-a)\cos(k_n a)}{k_n} + \frac{\sin(k_n a)}{k_n^2}.$$

The second line follows from the antiderivative $-(L-x)\cos(k_n x)/k_n - \sin(k_n x)/k_n^2$ and $\sin(k_n L) = 0$. Combining the projections,

$$A_n = \frac{2h}{L}\left(\frac{I_1}{a} + \frac{I_2}{L-a}\right)$$
$$= \frac{2h\sin(k_n a)}{Lk_n^2}\left(\frac{1}{a} + \frac{1}{L-a}\right) = \frac{2hL^2\sin(n\pi a/L)}{n^2\pi^2 a(L-a)}.$$

The $\cos(k_n a)$ terms cancel because the shape is continuous at the pluck point. At $a = L/2$, the first three nonzero coefficients have signs $+, -, +$; replacing all by positive magnitudes would not reconstruct a triangular shape.

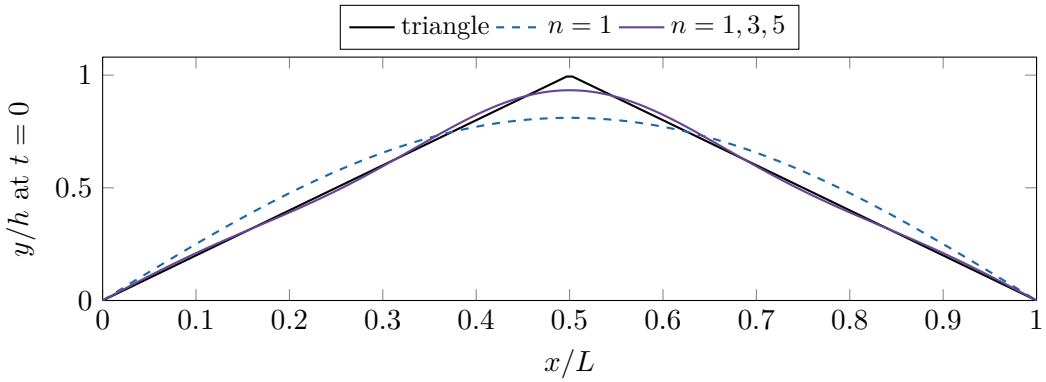


Figure 5.3: Successive odd Fourier components sharpen the midpoint triangle. Their signs and relative phases are essential, not just their magnitudes.

5.6.3 Derive the modal energy normalization

For $y = q_n(t) \sin k_n x$, the velocity is $\dot{q}_n \sin k_n x$ and the slope is $k_n q_n \cos k_n x$. Hence

$$T_n = \frac{\mu}{2}\dot{q}_n^2 \frac{L}{2} = \frac{\mu L}{4}\dot{q}_n^2, \quad U_n = \frac{\mathcal{T}}{2} k_n^2 q_n^2 \frac{L}{2} = \frac{\mu L}{4}\omega_n^2 q_n^2.$$

The effective modal mass in this unnormalized sine-coordinate convention is $\mu L/2$, since $T_n = M_n \dot{q}_n^2/2$. If the eigenfunction were rescaled, q_n and M_n would change inversely so that physical energy remains unchanged. Orthogonality of both sines and cosines makes the total energy a sum of independent positive modal energies.

Worked Example: An initial velocity with two spatial regions

Let $g(x) = V$ on $0 < x < L/2$ and $-V$ on $L/2 < x < L$, and let the initial shape be zero. The velocity coefficient is

$$B_n = \frac{2}{L\omega_n}\left[V\int_0^{L/2} \sin(k_n x)\mathrm{d}x - V\int_{L/2}^{L} \sin(k_n x)\mathrm{d}x\right] = \frac{2V}{n\pi\omega_n}[1 - 2\cos(n\pi/2) + (-1)^n].$$

For odd n the bracket is zero. For $n = 2 \pmod 4$, it is four, and for $n = 0 \pmod 4$, it is zero. Only $n = 2, 6, 10, \ldots$ occur. The antisymmetry about the midpoint predicts the absence of odd modes before integration. The endpoint velocities are interpreted as the interior limiting profile, as in the original rectangular-velocity example.

Standard Workflow: A standardized Fourier initial-value answer

1. List endpoint conditions and compatibility of $f(x), g(x)$ with fixed or free supports.
2. Derive $X_n(x), k_n, \omega_n$ from the spatial boundary problem.
3. State the normalization integral. Project f for cosine-in-time coefficients and g/ω_n for sine-in-time coefficients.
4. Split every piecewise integral at each change of formula. Use symmetry as an independent selection check.
5. Reconstruct $y(x,t)$ and verify $y(x,0)$ and $y_t(x,0)$.
6. Check modal energy against the direct initial energy integral when feasible.

5.7 Chapter Mind Map

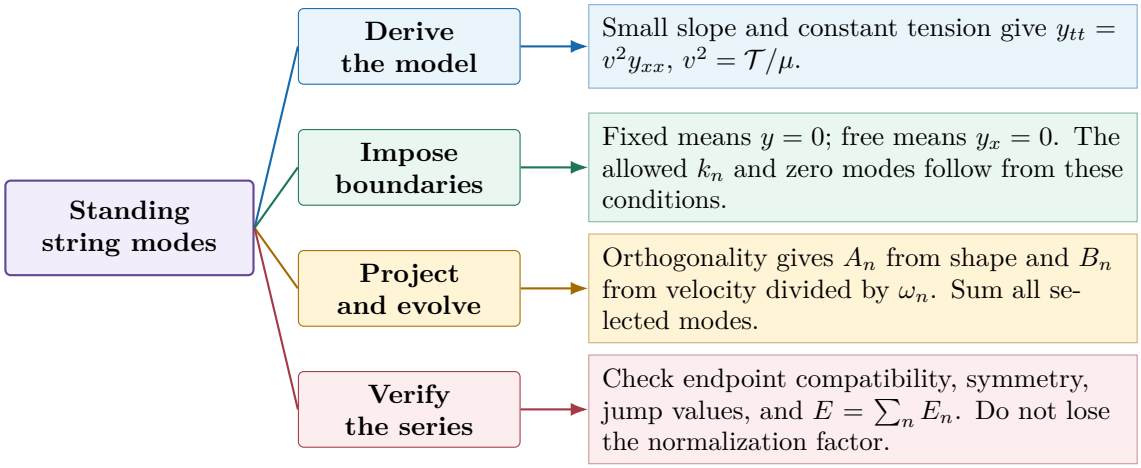


Figure 5.4: Chapter 5 mind map: boundaries determine the basis, and initial data determine its coefficients.

6 Traveling Waves, Energy Flux, and Scattering

Source map: Syllabus Unit 5; group-velocity connection to Peatross and Ware Section 7.2.

6.1 d'Alembert's solution and propagation direction

For $y_{tt} = v^2 y_{xx}$ on an infinite string, use $\xi = x - vt$, $\eta = x + vt$. The equation becomes $y_{\xi\eta} = 0$, hence

$$y(x,t) = F(x - vt) + G(x + vt).$$

A point of fixed argument $x - vt$ moves right; a fixed $x + vt$ moves left. A snapshot is not enough to determine propagation direction unless y_t is also supplied.

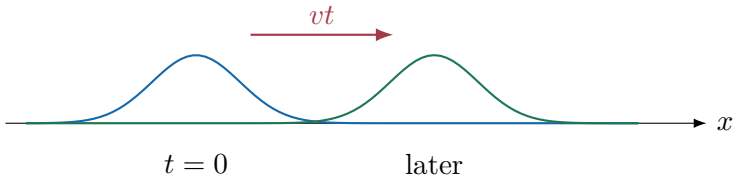


Figure 6.1: A right-moving pulse keeps its shape in a nondispersive medium.

Key Result: The infinite-string initial-value solution

If $y(x,0) = f(x)$ and $y_t(x,0) = g(x)$,

$$y(x,t) = \frac{f(x - vt) + f(x + vt)}{2} + \frac{1}{2v}\int_{x-vt}^{x+vt} g(s)\mathrm{d}s.$$

Derivation: $F + G = f$ and $-vF' + vG' = g$ give $F' = (f' - g/v)/2$, $G' = (f' + g/v)/2$. Integrating and substituting yields the formula. Differentiation at $t = 0$ checks both data.

Worked Example: A released pulse splits, but a launched pulse need not

If $g = 0$, a localized shape f separates into two pulses of half its height. To make a purely right-moving pulse $f(x - vt)$, choose $g(x) = -vf'(x)$. With $f = Ae^{-x^2/(2\sigma^2)}$, the required velocity is $g = vAxe^{-x^2/(2\sigma^2)}/\sigma^2$. Zero initial velocity and purely rightward propagation are incompatible for a

nonconstant pulse.

6.2 Energy density and directed power

The transverse kinetic-energy density is $u_K = \mu y_t^2/2$. For small slope the extra length of a segment is $\mathrm{d}\ell - \mathrm{d}x \simeq y_x^2 \mathrm{d}x/2$, giving $u_U = \mathcal{T} y_x^2/2$. Differentiate their sum and use the wave equation:

$$\begin{aligned}\partial_t u &= \mu y_t y_{tt} + \mathcal{T} y_x y_{xt} \\ &= \mathcal{T}(y_t y_{xx} + y_x y_{xt}) = \partial_x(\mathcal{T} y_t y_x).\end{aligned}$$

Key Result: Local conservation and string power

$$\partial_t u + \partial_x P = 0, \qquad P = -\mathcal{T} y_x y_t.$$

For $y = A\cos(kx - \omega t)$,

$$\langle P \rangle = \tfrac{1}{2}\mathcal{T} k\omega A^2 = \tfrac{1}{2} Z v_y^2 = \tfrac{1}{2} Z \omega^2 A^2, \qquad Z = \mathcal{T}/v = \sqrt{\mathcal{T}\mu}.$$

Here $v_y = \omega A$ denotes the peak transverse particle speed, not the propagation speed. The mechanical wave impedance Z has units $\mathrm{kg\,s^{-1}}$.

For a right-moving shape, $y_t = -v y_x$ and $P = vu \geq 0$; for a left-moving shape $P = -vu$. For a sum of opposite harmonic waves with complex displacement amplitudes A and B, cross terms cancel from the time-averaged directed flux:

$$\langle P \rangle = \tfrac{1}{2} Z \omega^2 (|A|^2 - |B|^2).$$

A standing wave can store energy while carrying zero net average flux.

Worked Example: Recover amplitude from power

For $\mathcal{T} = 40\,\mathrm{N}$, $\mu = 0.010\,\mathrm{kg\,m^{-1}}$, and $f = 20\,\mathrm{Hz}$, $v = 63.25\,\mathrm{m\,s^{-1}}$ and $Z = 0.6325\,\mathrm{kg\,s^{-1}}$. If $\langle P \rangle = 0.50\,\mathrm{W}$, then $A = \sqrt{2\langle P \rangle/(Z\omega^2)} = 0.0100\,\mathrm{m}$. The small-slope check is $kA \simeq 0.0199 \ll 1$; a small displacement relative to an arbitrary meter scale would not be the relevant condition.

6.3 An interface: derive amplitude coefficients from continuity

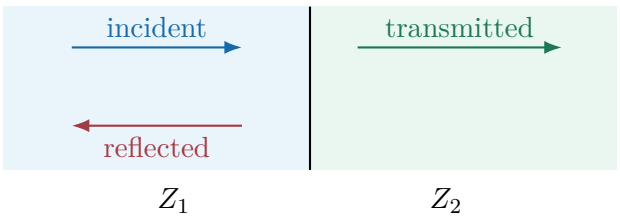


Figure 6.2: The same-frequency incident, reflected, and transmitted waves are matched at a stationary junction.

At $x = 0$, write $\tilde{y}_1 = A_i e^{\mathrm{i}k_1 x} + A_r e^{-\mathrm{i}k_1 x}$ and $\tilde{y}_2 = A_t e^{\mathrm{i}k_2 x}$. For a massless junction, displacement continuity and transverse-force balance give

$$A_i + A_r = A_t, \qquad \mathcal{T}_1 k_1 (A_i - A_r) = \mathcal{T}_2 k_2 A_t.$$

Because $\mathcal{T}_j k_j = \omega Z_j$, solution of these two linear equations yields

Key Result: Displacement and power coefficients

$$r = \frac{A_r}{A_i} = \frac{Z_1 - Z_2}{Z_1 + Z_2}, \quad t = \frac{A_t}{A_i} = \frac{2Z_1}{Z_1 + Z_2},$$

$$R = |r|^2, \quad T = \frac{Z_2}{Z_1}|t|^2 = \frac{4Z_1Z_2}{(Z_1+Z_2)^2}, \quad R+T=1.$$

This assumes real positive impedances and a lossless interface. For an ordinary collinear string junction, static tension must also be balanced; differing effective tensions require an appropriate support or waveguide model.

A heavier second string at the same tension has $Z_2 > Z_1$, so $r < 0$: reflection inverts the displacement. In the fixed-end limit $Z_2 \to \infty$, $r \to -1$. In the ideal free-end limit $Z_2 \to 0$, $r \to +1$ and $t \to 2$, yet transmitted power tends to zero. An amplitude greater than the incident amplitude does not violate energy conservation.

Worked Example: An amplitude coefficient is not an energy fraction

Let $\mu_2 = 4\mu_1$ at common tension. Then $Z_2 = 2Z_1$, $r = -1/3$, $t = 2/3$, $R = 1/9$, and $T = 8/9$. The tempting answer $T = 4/9$ fails because the second wave carries more power for a given displacement amplitude.

6.4 A localized barrier: a point mass

A mass M_b joins identical strings of impedance Z. Continuity gives $t = 1 + r$. The net vertical force on the mass is $\mathcal{T}[y_x(0^+) - y_x(0^-)]$. With $\mathrm{e}^{-\mathrm{i}\omega t}$,

$$-M_b\omega^2 A_t = 2\mathrm{i}\omega Z A_r.$$

Therefore

$$r = -\frac{M_b\omega}{M_b\omega + 2\mathrm{i}Z}, \quad t = \frac{2\mathrm{i}Z}{M_b\omega + 2\mathrm{i}Z}, \quad R = \frac{M_b^2\omega^2}{M_b^2\omega^2 + 4Z^2}, \quad T = 1 - R.$$

Low frequencies pass almost freely; high frequencies see an almost fixed boundary. A finite mass stores energy but does not dissipate it. Adding friction would change the energy balance.

Supplementary Background: A finite intermediate layer

Suppose equal exterior impedances Z_1 enclose a length d of impedance Z_2 and wavenumber k_2. Define $r = (Z_1 - Z_2)/(Z_1 + Z_2)$ and $\delta = k_2 d$. The initial transmitted amplitude is $(1 - r^2)\mathrm{e}^{\mathrm{i}\delta}$ times the incident amplitude; each additional round trip multiplies it by $r^2\mathrm{e}^{2\mathrm{i}\delta}$. Summing the geometric series gives

$$t_{\text{layer}} = \frac{(1-r^2)\mathrm{e}^{\mathrm{i}\delta}}{1 - r^2\mathrm{e}^{2\mathrm{i}\delta}}, \qquad T_{\text{layer}} = \frac{1}{1 + \frac{1}{4}(Z_2/Z_1 - Z_1/Z_2)^2 \sin^2\delta}.$$

The equal exterior impedances make $T = |t|^2$. Perfect transmission occurs at $k_2 d = n\pi$ even when either interface alone reflects. This is resonant cancellation of the total reflected wave, not an absence of internal reflections. It extends the syllabus barrier idea; Problem T8 gives the matching exercise.

6.5 Packets, phase speed, and group speed

For two nearly equal waves, trigonometric addition gives

$$\cos(k_1 x - \omega_1 t) + \cos(k_2 x - \omega_2 t) = 2\cos\frac{\Delta k x - \Delta\omega t}{2}\cos(\bar{k}x - \bar{\omega}t).$$

The carrier moves at $v_p \simeq \bar{\omega}/\bar{k}$ and the envelope at $v_g \simeq \Delta\omega/\Delta k$. For a narrow packet, expand $\omega(k) = \omega_c + v_g(k - k_c) + \frac{1}{2}\omega_c''(k - k_c)^2 + \cdots$. Retaining only the linear term translates the envelope without distortion at $v_g = \omega_c'$. Quadratic and higher terms reshape it.

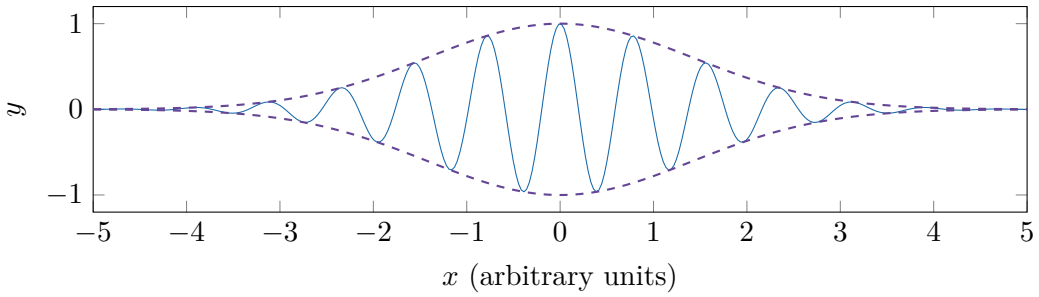


Figure 6.3: A carrier under a smooth envelope. The envelope speed comes from the slope of the dispersion relation.

Worked Example: A dispersive dispersion law

If $\omega = \alpha k^2$ with $k > 0$, then $v_p = \alpha k$ and $v_g = 2\alpha k$. A packet travels twice as fast as its crests to first order. Since $\omega'' = 2\alpha \neq 0$, a finite-bandwidth packet also spreads. A plot of $\omega(k)$ gives v_g from the tangent slope and v_p from the line to the origin.

Standard Workflow: Scattering questions

Write a direction-correct wave on each side; retain the same frequency at a stationary boundary; impose continuity and force balance including any junction mass; solve for complex amplitudes; convert to power using impedance ratios; check $R + T$ and the fixed/free/matched limits.

6.6 Derivation Workshop: Impedance and Barrier Matching

6.6.1 Fix the force convention before defining impedance

The transverse velocity is $v_y = y_t$. Define $F_{\rightarrow}$ as the signed vertical force (positive upward) exerted by the left part of the string on the right part at the cut, so $F_{\rightarrow} = -\mathcal{T} y_x$. For a right-moving wave, $y_t = -v y_x$, giving $F_{\rightarrow} = Z v_y$ with $Z = \mathcal{T}/v > 0$. For a left-moving wave, $F_{\rightarrow} = -Z v_y$. The signed power flowing through the cut toward $+x$ is $F_{\rightarrow} v_y = -\mathcal{T} y_x y_t$.

For harmonic displacement $\widetilde{y} = A\mathrm{e}^{\mathrm{i}kx} + B\mathrm{e}^{-\mathrm{i}kx}$,

$$\widetilde{v}_y = -\mathrm{i}\omega(A\mathrm{e}^{\mathrm{i}kx} + B\mathrm{e}^{-\mathrm{i}kx}), \quad \widetilde{F}_{\rightarrow} = -\mathrm{i}\omega Z(A\mathrm{e}^{\mathrm{i}kx} - B\mathrm{e}^{-\mathrm{i}kx}).$$

The product average $\frac{1}{2}\,\mathrm{Re}(\widetilde{F}_{\rightarrow}\widetilde{v}_y^*)$ has terms proportional to $|A|^2 - |B|^2$; its cross terms are purely imaginary and contribute no mean power. This proves the directed-flux formula for simultaneous counterpropagating waves.

6.6.2 A second derivation of finite-layer transmission

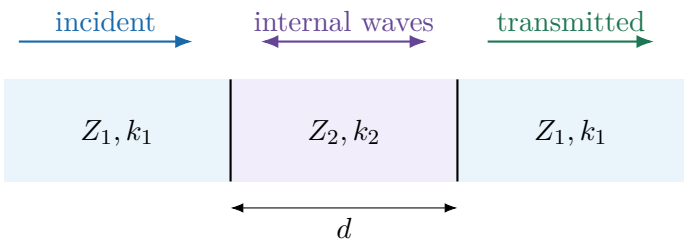


Figure 6.4: A finite layer requires both forward and backward internal waves. Equal exterior impedances make transmitted power proportional to $|t|^2$.

Within a uniform lossless segment define the state $\boldsymbol{s} = (\widetilde{y}, \widetilde{p})^T$ with $\widetilde{p} = \mathcal{T}\widetilde{y}_x$. From $\widetilde{y}'' + k^2\widetilde{y} = 0$,

propagation over d gives

$$\boldsymbol{s}(d) = \begin{pmatrix} \cos\delta & \sin\delta/(\omega Z_2) \\ -\omega Z_2 \sin\delta & \cos\delta \end{pmatrix} \boldsymbol{s}(0), \qquad \delta = k_2 d.$$

At the left edge, $\boldsymbol{s}(0) = (1 + r_{\text{layer}}, \mathrm{i}\omega Z_1[1 - r_{\text{layer}}])^T$. At the right edge, $\boldsymbol{s}(d) = (t_{\text{layer}}, \mathrm{i}\omega Z_1 t_{\text{layer}})^T$, where transmitted amplitude is referenced to $x = d$. Insert these expressions and eliminate r_{layer}:

$$t_{\text{layer}} = \frac{1}{\cos\delta - \frac{\mathrm{i}}{2}(Z_2/Z_1 + Z_1/Z_2)\sin\delta}.$$

Taking the squared modulus gives the same T as the internal-reflection sum in the original text. At $\delta = n\pi$, the propagation matrix is $(-1)^n I$: the layer changes only phase, producing perfect transmission. Two independent derivations agreeing is a strong check on interface signs.

6.6.3 The point barrier as a filter

The nondimensional parameter of the point-mass junction is $\xi = M_b\omega/(2Z)$. The derived coefficients become $r = -\xi/(\xi + \mathrm{i})$ and $t = \mathrm{i}/(\xi + \mathrm{i})$, so $R = \xi^2/(1 + \xi^2)$ and $T = 1/(1 + \xi^2)$. Half power is at $\xi = 1$. These formulas show that a heavy junction preferentially suppresses high-frequency transmitted displacement and power.

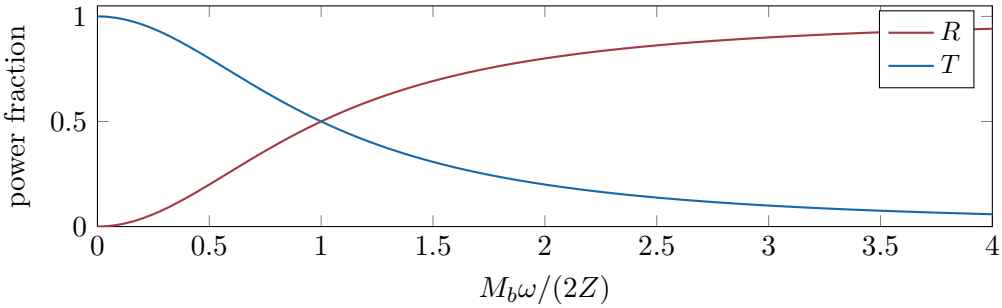


Figure 6.5: The point mass is lossless: reflection and transmission complement each other at every frequency.

Worked Example: Specify both transmitted magnitude and phase

For $M_b\omega = 2Z$, $t = \mathrm{i}/(1 + \mathrm{i}) = (1 + \mathrm{i})/2$ and $r = -(1 - \mathrm{i})/2$. Thus $|t| = |r| = 1/\sqrt{2}$, $T = R = 1/2$, and $\arg t = \pi/4$. With the convention $\mathrm{Re}(tA_i\mathrm{e}^{-\mathrm{i}\omega t})$, the junction displacement lags the incident displacement by $\pi/4$. Continuity is explicit: $1 + r = t$. Quoting only $T = 1/2$ would leave the requested displacement phase unanswered.

Standard Workflow: A reusable scattering answer

1. Draw each region, positive direction, interface location, and amplitude reference plane.
2. Write all allowed traveling or decaying terms; exclude a wave incident from the wrong infinity unless specified.
3. Use one common frequency at a stationary interface. Match displacement and force, adding junction inertia or dissipation if present.
4. Solve for complex r, t; retain phase information until the end.
5. Convert amplitudes to directed powers using the correct impedance factor.
6. Check continuity, $R + T = 1$ for a lossless system, and matched/free/fixed or low/high-frequency limits.

6.7 Chapter Mind Map

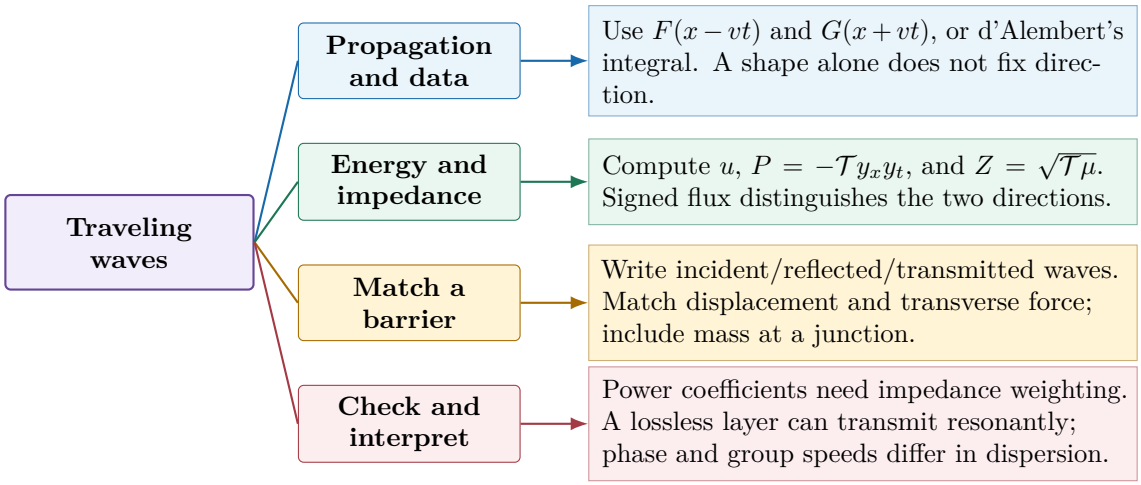


Figure 6.6: Chapter 6 mind map: amplitudes come from matching, while energy fractions come from directed flux.

7 Electromagnetic Waves and Energy Transport

Source map: Syllabus Unit 6; Peatross and Ware Sections 1.6–1.7, 2.1–2.3, 2.6–2.7.

7.1 Maxwell's equations and the material assumptions

For macroscopic fields,

$$\nabla\cdot\boldsymbol{D}=\rho_f, \qquad \nabla\cdot\boldsymbol{B}=0,$$
$$\nabla\times\boldsymbol{E}=-\partial_t\boldsymbol{B}, \qquad \nabla\times\boldsymbol{H}=\boldsymbol{J}_f+\partial_t\boldsymbol{D}.$$

In a homogeneous isotropic nondispersive linear material, $\boldsymbol{D}=\epsilon\boldsymbol{E}$ and $\boldsymbol{B}=\mu\boldsymbol{H}$. In optics we usually use nonmagnetic media with $\mu=\mu_0$. The polarization is $\boldsymbol{P}=\boldsymbol{D}-\epsilon_0\boldsymbol{E}$, its bound charge is $\rho_b=-\nabla\cdot\boldsymbol{P}$, and its current is $\boldsymbol{J}_b=\partial_t\boldsymbol{P}$. Material polarization and the polarization state of a wave are different meanings of the same word.

7.2 Derive the wave equation, including the source terms

In vacuum, take the curl of Faraday's law and use Ampere's law:

$$\nabla\times(\nabla\times\boldsymbol{E})=-\mu_0\epsilon_0\partial_t^2\boldsymbol{E}-\mu_0\partial_t\boldsymbol{J}.$$

The curl-curl identity and Gauss's law give

$$\nabla^2\boldsymbol{E}-\frac{1}{c^2}\partial_t^2\boldsymbol{E}=\mu_0\partial_t\boldsymbol{J}+\frac{1}{\epsilon_0}\nabla\rho, \qquad c=\frac{1}{\sqrt{\mu_0\epsilon_0}}.$$

If $\rho_f=0$ and $\boldsymbol{J}=\boldsymbol{J}_f+\partial_t\boldsymbol{P}$, then

$$\nabla^2\boldsymbol{E}-\frac{1}{c^2}\boldsymbol{E}_{tt}=\mu_0\partial_t\boldsymbol{J}_f+\mu_0\boldsymbol{P}_{tt}-\frac{1}{\epsilon_0}\nabla(\nabla\cdot\boldsymbol{P}).$$

In a source-free homogeneous isotropic material, $\nabla\cdot\boldsymbol{D}=0$ implies $\nabla\cdot\boldsymbol{E}=0$, and the nondispersive equation reduces to $\nabla^2\boldsymbol{E}=\mu\epsilon\boldsymbol{E}_{tt}$. In an inhomogeneous or anisotropic material, $\nabla\cdot\boldsymbol{E}=0$ cannot simply be assumed.

Key Result: Plane-wave constraints are stronger than the wave equation

For $\boldsymbol{E} = \mathrm{Re}[\boldsymbol{E}_0 \mathrm{e}^{\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)}]$ in a source-free isotropic material,

$$k^2 = \mu\epsilon\omega^2, \quad \boldsymbol{k}\cdot\boldsymbol{E}_0 = 0, \quad \boldsymbol{B}_0 = \frac{\boldsymbol{k}\times\boldsymbol{E}_0}{\omega}, \quad \boldsymbol{H}_0 = \frac{1}{Z}\hat{\boldsymbol{k}}\times\boldsymbol{E}_0,$$

$$v_p = \frac{1}{\sqrt{\mu\epsilon}}, \qquad Z = \sqrt{\frac{\mu}{\epsilon}}.$$

The last formula assumes a real propagation direction in a lossless material. In vacuum, $E_0 = cB_0$, $Z_0 \simeq 376.73\,\Omega$. The wave equation alone permits unphysical longitudinal field choices unless Maxwell's divergence constraints are imposed too.

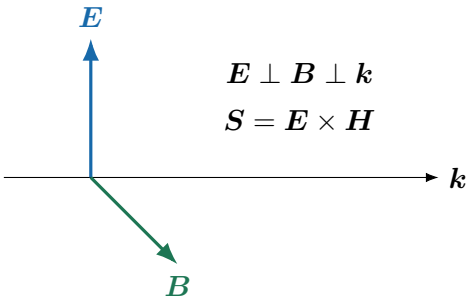


Figure 7.1: The right-handed field relation fixes the direction of energy transport.

Worked Example: Find the accompanying magnetic field correctly

In vacuum, $\boldsymbol{E} = E_0\cos(kz-\omega t)\hat{\boldsymbol{x}}$. Then $\boldsymbol{B} = (E_0/c)\cos(kz-\omega t)\hat{\boldsymbol{y}}$, since $\hat{\boldsymbol{z}}\times\hat{\boldsymbol{x}} = \hat{\boldsymbol{y}}$. Reversing propagation while keeping the electric polarization makes $\boldsymbol{B}$ reverse sign. A magnetic field in $-\hat{\boldsymbol{y}}$ with the original forward phase would send $\boldsymbol{S}$ backward and violate Faraday's law.

7.3 Poynting's theorem from Maxwell's equations

Dot Ampere's law with $\boldsymbol{E}$ and Faraday's law with $\boldsymbol{H}$, then use the divergence identity:

$$\nabla\cdot(\boldsymbol{E}\times\boldsymbol{H}) = -\boldsymbol{E}\cdot\boldsymbol{J}_f - \boldsymbol{E}\cdot\partial_t\boldsymbol{D} - \boldsymbol{H}\cdot\partial_t\boldsymbol{B}.$$

For real, time-independent, nondispersive ϵ and μ, the last two terms are time derivatives, so

Key Result: Local and integral energy conservation

$$\partial_t u + \nabla\cdot\boldsymbol{S} = -\boldsymbol{J}_f\cdot\boldsymbol{E}, \quad u = \tfrac{1}{2}\epsilon E^2 + \tfrac{1}{2\mu}B^2, \quad \boldsymbol{S} = \boldsymbol{E}\times\boldsymbol{H}.$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_V u\,\mathrm{d}V + \oint_{\partial V}\boldsymbol{S}\cdot\hat{\boldsymbol{n}}\,\mathrm{d}A = -\int_V \boldsymbol{J}_f\cdot\boldsymbol{E}\,\mathrm{d}V.$$

Positive outward flux removes field energy; positive $\boldsymbol{J}_f\cdot\boldsymbol{E}$ transfers energy to charges. With polarization treated explicitly, vacuum field energy plus the medium's mechanical energy gives the appropriate balance.

For harmonic fields the exact cycle-average identity is

$$\langle\boldsymbol{S}\rangle = \tfrac{1}{2}\,\mathrm{Re}(\widetilde{\boldsymbol{E}}\times\widetilde{\boldsymbol{H}}^{*}).$$

For a single lossless plane wave,

$$I = \frac{|\boldsymbol{E}_0|^2}{2Z} = \tfrac{1}{2}\epsilon v_p|\boldsymbol{E}_0|^2, \quad \langle u\rangle = \tfrac{1}{2}\epsilon|\boldsymbol{E}_0|^2, \quad I = v_p\,\langle u\rangle\,.$$

In nonmagnetic material, $Z = Z_0/n$ and $I = \frac{1}{2}n\epsilon_0 c|\boldsymbol{E}_0|^2$. A complex vector norm is $\boldsymbol{E}_0 \cdot \boldsymbol{E}_0^*$; $\boldsymbol{E}_0 \cdot \boldsymbol{E}_0$ is not an intensity.

Worked Example: A beam's electric and magnetic amplitudes

A vacuum beam has power $P = 2.0\,\mathrm{mW}$ uniformly spread over $A = 1.0\,\mathrm{mm}^2$. Thus $I = 2000\,\mathrm{W\,m^{-2}}$, $E_0 = \sqrt{2Z_0 I} = 1.228 \times 10^3\,\mathrm{V\,m^{-1}}$, $B_0 = E_0/c = 4.095\,\mu\mathrm{T}$, and $\langle u \rangle = I/c = 6.671 \times 10^{-6}\,\mathrm{J\,m^{-3}}$. For a Gaussian beam, the stated area would need a specified averaging convention; uniform illumination is part of this model.

7.4 Absorption and complex wave impedance

Use a forward wave with $\tilde{n} = n + \mathrm{i}\kappa$, $\tilde{k} = (\omega/c)(n + \mathrm{i}\kappa)$:

$$\widetilde{E}(z) = E_0 \mathrm{e}^{\mathrm{i}n\omega z/c}\mathrm{e}^{-\kappa\omega z/c}.$$

In a nonmagnetic material $\widetilde{H} = \tilde{n}\widetilde{E}/Z_0$ (with the appropriate perpendicular direction), so E and H are generally out of phase. Their mean flux is

$$I(z) = \frac{n}{2Z_0}|E_0|^2\mathrm{e}^{-2\kappa\omega z/c} = I(0)\mathrm{e}^{-\alpha z}, \qquad \alpha = \frac{2\kappa\omega}{c} = \frac{4\pi\kappa}{\lambda_0}.$$

The amplitude penetration length is $\delta_A = c/(\kappa\omega)$; the intensity 1/e length is $\delta_I = 1/\alpha = \delta_A/2$.

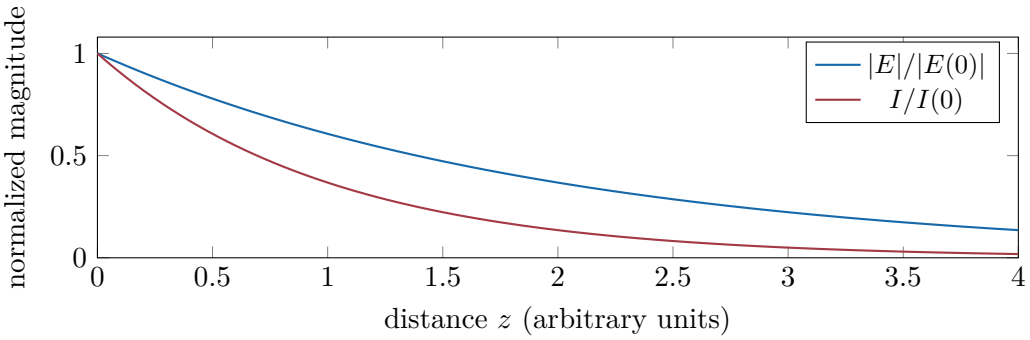


Figure 7.2: An intensity attenuation exponent is twice the field-amplitude exponent.

Worked Example: Absorption measured through two thicknesses

If otherwise identical samples of thicknesses d_1 and d_2 transmit internal intensities I_1 and I_2, and common interface factors cancel, then $\alpha = \ln(I_1/I_2)/(d_2 - d_1)$. For $d_2 - d_1 = 2.0\,\mathrm{mm}$ and $I_2/I_1 = 0.60$, $\alpha = 255.4\,\mathrm{m^{-1}}$. At $\lambda_0 = 600\,\mathrm{nm}$, $\kappa = \alpha\lambda_0/(4\pi) = 1.220 \times 10^{-5}$. Using the external incident intensity without correcting reflection would generally overestimate absorption.

7.5 A useful boundary connection

At normal incidence between lossless dielectrics, tangential E and H continuity give

$$1 + r_E = t_E, \qquad \frac{1 - r_E}{Z_1} = \frac{t_E}{Z_2},$$

$$r_E = \frac{Z_2 - Z_1}{Z_1 + Z_2}, \quad t_E = \frac{2Z_2}{Z_1 + Z_2}, \quad T = \frac{Z_1}{Z_2}|t_E|^2.$$

This is a supplementary bridge to the string interface: the power weighting is inverted because electromagnetic power is $E^2/(2Z)$ whereas string power is $Z\omega^2A^2/2$. It is not safe to copy a string displacement coefficient into an electric-field formula.

Error Check: Energy in a dispersive material

Do not use $u = (\epsilon E^2 + B^2/\mu)/2$ with a complex frequency-dependent ϵ as if it were an instantaneous stored energy. Dispersion introduces material oscillator energy, and loss requires explicit dissipation. Chapter 8 derives the Lorentz-medium balance. For a narrowband wave in a transparent dispersive medium, the cycle-averaged energy is $[\mathrm{d}(\omega\epsilon)/\mathrm{d}\omega]|E_0|^2/4 + \mu_0|H_0|^2/4$ under the stated nonmagnetic, low-loss assumptions.

7.6 Derivation Workshop: Reconstructing a Physical Plane Wave

7.6.1 Apply all four Maxwell equations in Fourier space

For a homogeneous isotropic medium with no free charges or currents, insert $\mathrm{e}^{\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)}$ into Maxwell's equations. With $\boldsymbol{D}_0 = \epsilon\boldsymbol{E}_0$, $\boldsymbol{B}_0 = \mu\boldsymbol{H}_0$, the four algebraic relations are

$$\begin{aligned} \boldsymbol{k}\cdot\epsilon\boldsymbol{E}_0 &= 0, & \boldsymbol{k}\cdot\boldsymbol{B}_0 &= 0, \\ \boldsymbol{k}\times\boldsymbol{E}_0 &= \omega\boldsymbol{B}_0, & \boldsymbol{k}\times\boldsymbol{B}_0 &= -\mu\epsilon\omega\boldsymbol{E}_0. \end{aligned}$$

For nonzero scalar ϵ, the first gives $\boldsymbol{k}\cdot\boldsymbol{E}_0 = 0$. Cross the third relation with $\boldsymbol{k}$ and use the fourth:

$$\boldsymbol{k}\times(\boldsymbol{k}\times\boldsymbol{E}_0) = \boldsymbol{k}(\boldsymbol{k}\cdot\boldsymbol{E}_0) - k^2\boldsymbol{E}_0 = -\mu\epsilon\omega^2\boldsymbol{E}_0.$$

A nonzero transverse field therefore requires $k^2 = \mu\epsilon\omega^2$. This derivation obtains transversality, dispersion, and the field ratio together. It is stronger than substituting a cosine into a scalar wave equation.

Error Check: What is special about a zero constitutive coefficient?

Dividing $\boldsymbol{k}\cdot\epsilon\boldsymbol{E}_0 = 0$ by ϵ assumes $\epsilon \neq 0$. At a material zero or in a strongly nonlocal model, longitudinal material modes may require separate treatment. Such plasma-mode analysis is not part of the present syllabus derivation. In an anisotropic crystal, $\boldsymbol{D}$ is transverse but $\boldsymbol{E}$ need not be; do not extend the scalar-medium result silently.

7.6.2 Complex impedance and the real part of the power

For a forward wave in a nonmagnetic medium, choose electric polarization along x and propagation along z. Then $\widetilde{H}_y = \tilde{n}\widetilde{E}_x/Z_0$ and the electric-to-magnetic impedance is $\widetilde{Z} = Z_0/\tilde{n}$. Its phasor relation is complex, but the transported mean power is real:

$$\langle S_z\rangle = \frac{1}{2}\,\mathrm{Re}(\widetilde{E}_x\widetilde{H}_y^*) = \frac{|\widetilde{E}_x|^2}{2}\,\mathrm{Re}\left(\frac{1}{\widetilde{Z}^*}\right) = \frac{n}{2Z_0}|\widetilde{E}_x|^2.$$

In particular, $|E|^2/(2|\widetilde{Z}|)$ is generally not the correct real power in a lossy medium. The relative phase of E and H matters.

Worked Example: A nonzero field magnitude with reduced real power

At one point let $\tilde{n} = 1 + \mathrm{i}$, and take a peak electric amplitude E_a. Then $|\widetilde{H}| = \sqrt{2}E_a/Z_0$, but the mean normal power is $E_a^2/(2Z_0)$ rather than $\sqrt{2}E_a^2/(2Z_0)$. The extra magnetic magnitude is partly in quadrature with the electric field. The direct phasor cross product gives the correct result without an ambiguous "effective impedance magnitude."

7.6.3 Standing-wave energy as a local conservation test

The original counterpropagating-wave sum gives

$$u(z,t) = 2\epsilon_0 E_a^2[\cos^2 kz\cos^2\omega t + \sin^2 kz\sin^2\omega t] = \epsilon_0 E_a^2[1 + \cos 2kz\cos 2\omega t].$$

Using $S_z = \epsilon_0 c E_a^2 \sin 2kz \sin 2\omega t$, differentiation gives

$$u_t = -2\epsilon_0 E_a^2 \omega \cos 2kz \sin 2\omega t, \quad (S_z)_z = 2\epsilon_0 c E_a^2 k \cos 2kz \sin 2\omega t.$$

They cancel because $\omega = ck$. This explicitly checks the instantaneous conservation law, not only its zero time average.

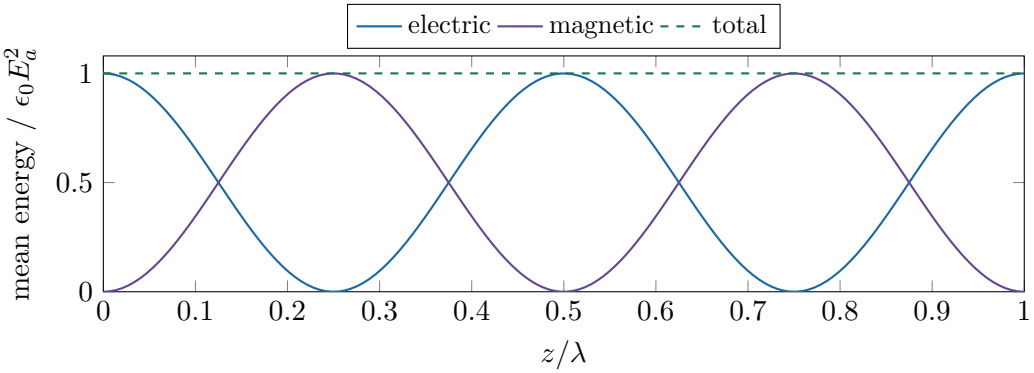


Figure 7.3: Electric and magnetic mean energies alternate spatially. Their sum is uniform although their instantaneous sum oscillates locally.

Standard Workflow: The standard electromagnetic field route

1. State the medium: vacuum or material; isotropic or anisotropic; lossless or absorbing; nondispersive or frequency-dependent.
2. Choose the propagation direction and phasor convention. Use the divergence law to test the proposed polarization.
3. Obtain $\boldsymbol{B}$ from $\boldsymbol{k} \times \boldsymbol{E}/\omega$ and then $\boldsymbol{H}$ from the constitutive relation.
4. Check the other curl equation and the resulting dispersion relation.
5. For power, use $\frac{1}{2}\,\mathrm{Re}(\widetilde{\boldsymbol{E}} \times \widetilde{\boldsymbol{H}}^*)$; for a detector surface, take its normal component.
6. For energy, distinguish nondispersive field storage from dispersive material storage. Check the Poynting balance and absorbed power sign.

7.7 Chapter Mind Map

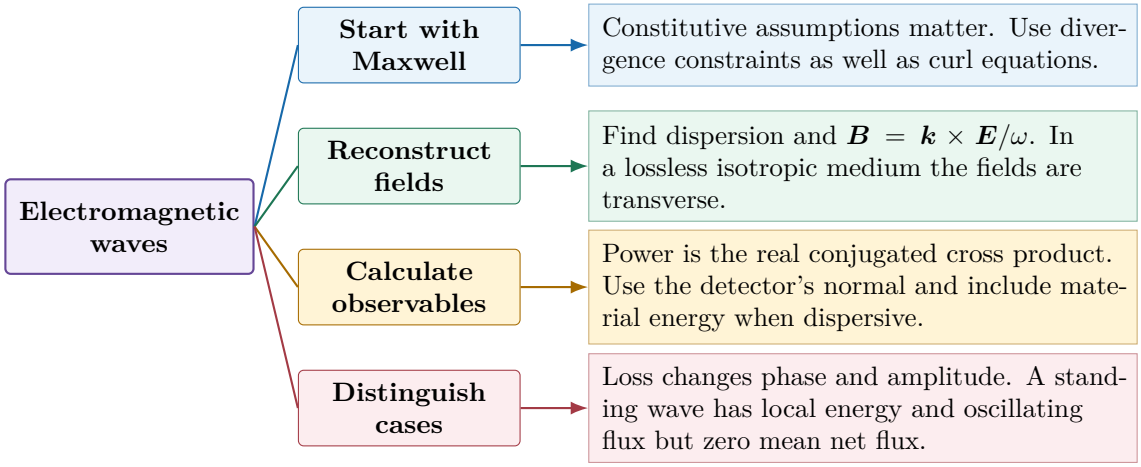


Figure 7.4: Chapter 7 mind map: a physical electromagnetic wave must satisfy field constraints and energy conservation together.

8 The Lorentz Model: From a Driven Electron to Optical Response

Source map: Syllabus Unit 7; Peatross and Ware Sections 2.3–2.5.

8.1 Microscopic displacement and macroscopic polarization

Let an electron of charge $-e$ and mass m_e have displacement x relative to its positive binding center. The linear model is

$$m_e\ddot{x} + m_e\Gamma\dot{x} + m_e\Omega_0^2 x = -eE(t).$$

The light field is treated as uniform across an atom, the displacement remains small, and magnetic-force effects are negligible for nonrelativistic electron motion. Damping Γ is the coefficient of $\dot{x}$, so it corresponds to 2β in the mechanical chapter. With N identical active oscillators per volume, the dipole density is $P = N(-e)x$.

Key Result: Derive the complex susceptibility

For $E = \mathrm{Re}(E_0 e^{-i\omega t})$,

$$X = -\frac{eE_0/m_e}{\Omega_0^2 - \omega^2 - i\Gamma\omega}, \quad P_0 = \epsilon_0\chi E_0,$$

$$\chi(\omega) = \frac{\Omega_p^2}{\Omega_0^2 - \omega^2 - i\Gamma\omega}, \qquad \Omega_p^2 = \frac{Ne^2}{\epsilon_0 m_e}.$$

The charge appears twice, once in the force and once in the dipole, giving a positive e^2 numerator. The susceptibility is dimensionless.

Let $\Delta = \Omega_0^2 - \omega^2$. Rationalizing gives

$$\chi' = \frac{\Omega_p^2\Delta}{\Delta^2 + \Gamma^2\omega^2}, \qquad \chi'' = \frac{\Omega_p^2\Gamma\omega}{\Delta^2 + \Gamma^2\omega^2}.$$

For a passive oscillator and $\omega > 0$, $\chi'' \geq 0$. The real part controls the in-phase material response; the imaginary part gives the quadrature response that absorbs energy.

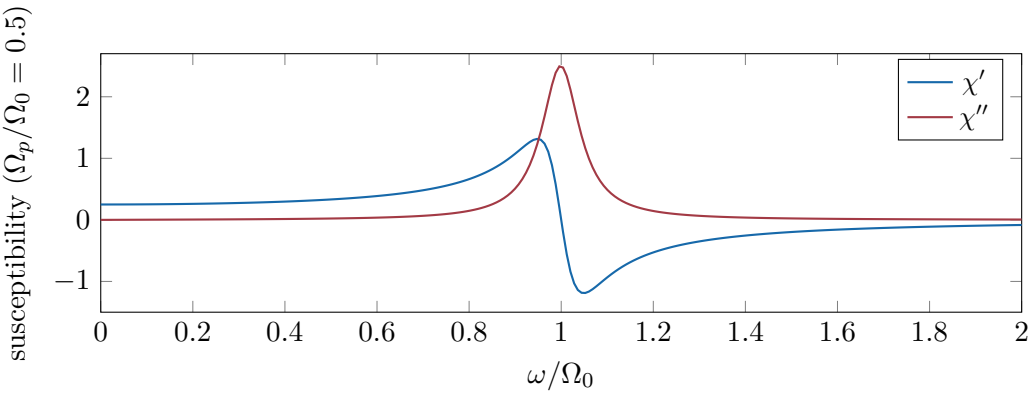


Figure 8.1: Real and imaginary susceptibility for $\Gamma/\Omega_0 = 0.1$. The real part changes sign near the resonance, while the absorptive part is positive.

8.2 From susceptibility to index, phase speed, and absorption

For an isotropic nonmagnetic material, Maxwell's equation gives

$$\widetilde{k}^2 = \frac{\omega^2}{c^2}(1+\chi), \qquad (n+\mathrm{i}\kappa)^2 = 1+\chi'+\mathrm{i}\chi''.$$

Equating real and imaginary parts gives $n^2-\kappa^2 = 1+\chi'$ and $2n\kappa = \chi''$. Let $A = 1+\chi'$, $B = \chi''$, and $R = \sqrt{A^2+B^2}$. Then

Key Result: Exact passive square root and observable quantities

$$n = \sqrt{\frac{R+A}{2}}, \qquad \kappa = \sqrt{\frac{R-A}{2}} \quad (B \geq 0),$$

$$v_p = \frac{c}{n}, \quad \lambda = \frac{\lambda_0}{n}, \quad \alpha = \frac{2\omega\kappa}{c}.$$

When $n > 0$, $\kappa = B/(2n)$ is equivalent. The square-root form also handles $n = 0$ without dividing by zero. If $|\chi| \ll 1$, $n \simeq 1+\chi'/2$ and $\kappa \simeq \chi''/2$. Small absorption alone does not imply small χ'.

Worked Example: Compute an index without a weak-response approximation

Suppose $\chi' = 1.24$ and $\chi'' = 0.30$. Then $A = 2.24$, $R = 2.26$, so $n = 1.50$ and $\kappa = 0.10$. At $\lambda_0 = 600\,\mathrm{nm}$, $\lambda = 400\,\mathrm{nm}$, $\delta_A = \lambda_0/(2\pi\kappa) = 955\,\mathrm{nm}$, and $\delta_I = 477\,\mathrm{nm}$. Checking $2n\kappa = 0.30$ catches the sign and factor-of-two conventions.

8.3 Energy transfer explains the imaginary part

The polarization current amplitude is $J_{P,0} = -\mathrm{i}\omega P_0$. Therefore

$$\langle E J_P \rangle = \frac{1}{2}\,\mathrm{Re}[E_0 J_{P,0}^*] = \frac{1}{2}\epsilon_0\omega\chi''|E_0|^2 \geq 0.$$

Independently, each electron dissipates $m_e\Gamma\langle \dot{x}^2\rangle = m_e\Gamma\omega^2|X|^2/2$. Multiplying by N and inserting X gives the same expression. This verifies both the microscopic model and the phasor sign.

Multiply the microscopic equation by $N\dot{x}$:

$$\partial_t\left[\frac{Nm_e}{2}(\dot{x}^2+\Omega_0^2x^2)\right] = E\dot{P} - Nm_e\Gamma\dot{x}^2.$$

Combining with vacuum-field Poynting conservation yields

$$\partial_t(u_{\mathrm{vac}}+u_{\mathrm{osc}}) + \nabla\cdot\boldsymbol{S} = -Nm_e\Gamma\dot{x}^2.$$

The missing energy becomes heat (or other reservoir energy represented by damping); it has not disappeared. In a monochromatic homogeneous wave, the flux decrease obeys $-\mathrm{d}I/\mathrm{d}z = \alpha I$. Since $\alpha I = (2\omega\kappa/c)(n\epsilon_0 c|E|^2/2) = \epsilon_0\omega\chi''|E|^2/2$, the propagation and oscillator calculations agree exactly.

Worked Example: At the binding resonance

For $\omega = \Omega_0$, $\chi' = 0$ and $\chi'' = \Omega_p^2/(\Gamma\Omega_0)$. This does not force $n = 1$: the exact square root includes χ''^2. Only in the dilute weak-response limit is $n \simeq 1$. Absorbed power has a maximum at Ω_0 for this damping model, but the peak of χ'' itself is slightly shifted because power contains the extra factor ω. Distinguish the measured response before differentiating.

8.4 Low, high, and resonant frequency limits

At $\omega \ll \Omega_0$, $\chi' \simeq \Omega_p^2/\Omega_0^2 > 0$ and $\chi'' \simeq \Omega_p^2\Gamma\omega/\Omega_0^4$. Far above resonance, $\chi' \simeq -\Omega_p^2/\omega^2$ and $\chi'' \simeq \Omega_p^2\Gamma/\omega^3$, so $\tilde{n} \to 1$. Near resonance, assuming $|\delta|, \Gamma \ll \Omega_0$ with $\delta = \omega - \Omega_0$,

$$\chi \simeq -\frac{\Omega_p^2}{2\Omega_0}\frac{1}{\delta + \mathrm{i}\Gamma/2}.$$

This produces the narrow absorptive line and rapid real-index variation. Normal dispersion means $\mathrm{d}n/\mathrm{d}\omega > 0$; anomalous dispersion means it is negative. Near significant absorption a group velocity computed from a real-index derivative may describe peak reshaping, not a signal or energy velocity.

Supplementary Background: Free-electron and multi-resonance connections

Setting $\Omega_0 = 0$ gives the Drude form $\chi = -\Omega_p^2/(\omega^2 + \mathrm{i}\Gamma\omega)$. In the collisionless limit, $n^2 = 1 - \Omega_p^2/\omega^2$. Above Ω_p waves propagate; below it the bulk solution is evanescent. A decaying evanescent field in a lossless plasma is not dissipative absorption: $\chi'' = 0$ and there is no heating. Multiple bound species contribute $\chi = \sum_j f_j\Omega_{pj}^2/(\Omega_j^2 - \omega^2 - \mathrm{i}\Gamma_j\omega)$. Local-field corrections and microscopic quantum derivations are outside the supplied syllabus.

Worked Example: A transparent plasma separates phase and group speeds

From $\omega^2 = \Omega_p^2 + c^2k^2$, $v_p = \omega/k = c/n$ and $v_g = c^2k/\omega = cn$, hence $v_pv_g = c^2$. At $\omega = 2\Omega_p$, $n = \sqrt{3}/2$, $v_p = 2c/\sqrt{3} > c$, and $v_g = \sqrt{3}c/2 < c$. A superluminal crest speed is not superluminal information transport.

Standard Workflow: Material-response questions

State the phasor convention; solve the microscopic driven equation; multiply by charge and density; separate χ' and χ''; choose the passive square root; calculate wavelength, phase speed, and intensity attenuation; verify $2n\kappa = \chi''$ and positive absorbed power. Use a weak-response approximation only after checking its small parameter.

8.5 Derivation Workshop: Index, Dissipation, and Stored Energy

8.5.1 Solve the complex square root without a hidden approximation

Write $A = 1 + \chi'$, $B = \chi''$, and $R = \sqrt{A^2 + B^2}$. From $n^2 - \kappa^2 = A$ and $2n\kappa = B$,

$$(n^2 + \kappa^2)^2 = (n^2 - \kappa^2)^2 + (2n\kappa)^2 = A^2 + B^2.$$

Since $n^2 + \kappa^2 \geq 0$, it equals R. Adding and subtracting with $n^2 - \kappa^2 = A$ gives $n^2 = (R + A)/2$ and $\kappa^2 = (R - A)/2$. The passive branch for an ordinary nonmagnetic medium at positive frequency has $n, \kappa \geq 0$. If $B = 0$ and $A < 0$, then $n = 0$ and $\kappa = \sqrt{-A}$: the field is evanescent, but not necessarily dissipative. To decide heating, inspect χ'', not only κ.

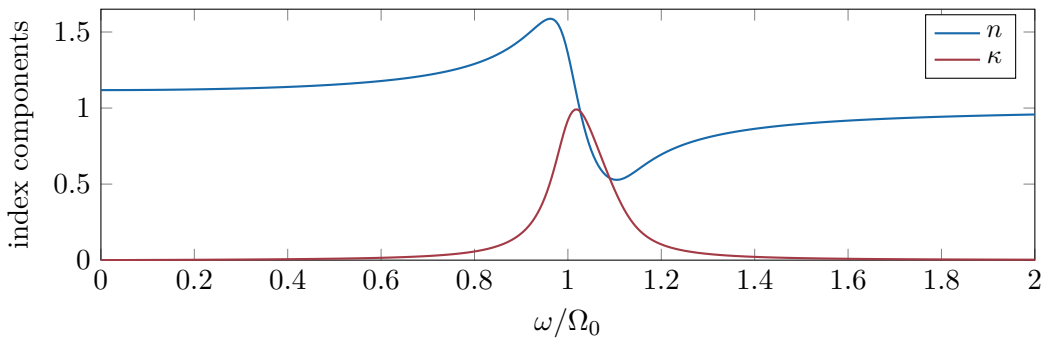


Figure 8.2: The exact n and κ generated by the same Lorentz parameters as the susceptibility plot: $\Omega_p/\Omega_0 = 0.5$, $\Gamma/\Omega_0 = 0.1$. Index variation and attenuation arise from one response function.

8.5.2 Separate three distinct approximation tests

- **Small loss:** $\kappa \ll n$ can justify neglecting κ^2 relative to n^2, leading to $n \simeq \sqrt{1+\chi'}$ when $1+\chi' > 0$. It does not make n close to one.
- **Weak susceptibility:** $|\chi| \ll 1$ permits $\widetilde{n} \simeq 1 + \chi/2$. This is a stronger and different statement.
- **Narrow resonance:** $\Gamma \ll \Omega_0$ and $|\omega - \Omega_0| \ll \Omega_0$ permit the local Lorentz-line approximation. They do not imply weak susceptibility at the resonance.

The strength, damping, detuning, and propagation length all affect the observable. A small κ can still give substantial attenuation when $\alpha d = 2\omega\kappa d/c$ is not small.

8.5.3 Derive the transparent dispersive energy formula

Consider the lossless single-oscillator model away from its resonance. With $\Delta = \Omega_0^2 - \omega^2$,

$$\chi = \frac{\Omega_p^2}{\Delta}, \quad |X|^2 = \frac{e^2|E_0|^2}{m_e^2\Delta^2}, \quad \langle u_{\text{osc}}\rangle = \frac{\epsilon_0\Omega_p^2(\omega^2+\Omega_0^2)|E_0|^2}{4\Delta^2}.$$

Differentiate the constitutive response:

$$\frac{\mathrm{d}(\omega\epsilon)}{\mathrm{d}\omega} = \epsilon_0\left[1 + \frac{\Omega_p^2}{\Delta} + \frac{2\Omega_p^2\omega^2}{\Delta^2}\right] = \epsilon_0\left[1 + \frac{\Omega_p^2(\Omega_0^2+\omega^2)}{\Delta^2}\right].$$

Therefore the cycle-averaged vacuum electric energy $\epsilon_0|E_0|^2/4$ plus oscillator energy is $\frac{1}{4}[\mathrm{d}(\omega\epsilon)/\mathrm{d}\omega]|E_0|^2$. Including magnetic energy gives the transparent, nonmagnetic, narrowband storage formula

$$\overline{u}_{\text{total}} = \frac{1}{4}\frac{\mathrm{d}(\omega\epsilon)}{\mathrm{d}\omega}|E_0|^2 + \frac{\mu_0}{4}|H_0|^2.$$

This proof identifies the missing material energy directly; it does not assume that a complex ϵ is a time-local energy coefficient.

For a propagating transparent wave, $\epsilon = \epsilon_0 n^2$ and $|H_0| = n|E_0|/Z_0$. Then

$$\overline{u}_{\text{total}} = \frac{\epsilon_0}{2}n(n+\omega n')|E_0|^2, \quad I = \frac{\epsilon_0 c}{2}n|E_0|^2, \quad \frac{I}{\overline{u}_{\text{total}}} = \frac{c}{n+\omega n'} = v_g.$$

Thus energy velocity equals group velocity in this transparent local model under the stated assumptions. This equality cannot be transplanted unchanged into a strongly absorbing, reshaping pulse problem.

Worked Example: Check the energy interpretation in a transparent plasma

For $n^2 = 1 - \Omega_p^2/\omega^2 > 0$, differentiation gives $n_g = n + \omega n' = 1/n$. Hence $nn_g = 1$ and $\overline{u}_{\text{total}} = \epsilon_0|E_0|^2/2$. The power is $n\epsilon_0 c|E_0|^2/2$, so energy speed is $nc = v_g < c$. Replacing the total energy by the naive nondispersive expression would instead give the incorrect speed c/n.

Standard Workflow: The standard material-response calculation

1. Write the electron equation and identify the damping coefficient: Γ here equals 2β in the mechanical notation.
2. Solve for X; use both charge factors in $P = N(-e)X$ to obtain χ.
3. Separate real and imaginary parts. Test passivity through $\omega\chi'' \geq 0$.
4. Solve the exact complex square root first; simplify only after testing an explicit small parameter.
5. Report $\lambda = \lambda_0/n$, $v_p = c/n$, and $\alpha = 2\omega\kappa/c$ with amplitude/intensity lengths distinguished.
6. If asked about energy, compute $\left\langle E\dot{P}\right\rangle$ or the total field-plus-oscillator balance. Frequency

derivatives are required for group velocity.

8.6 Chapter Mind Map

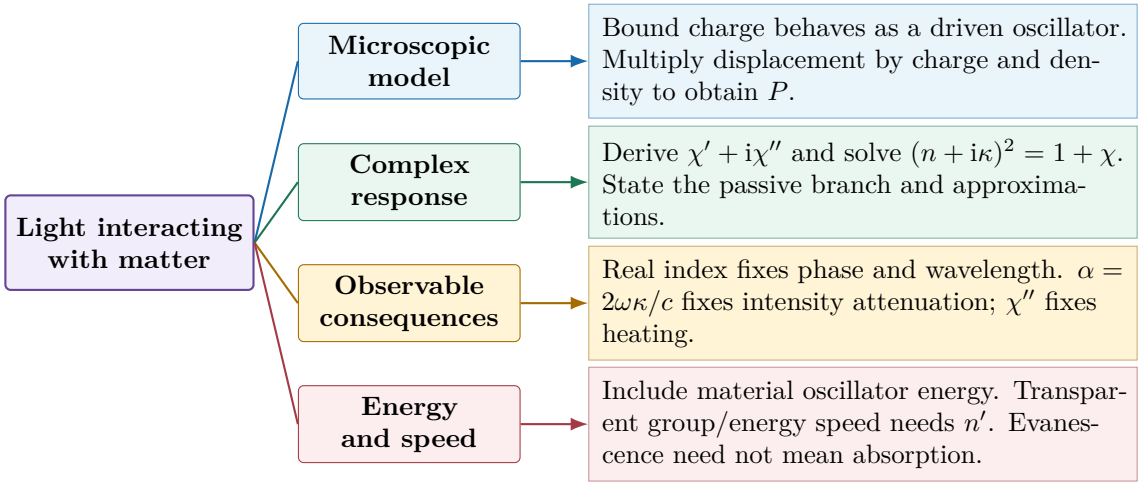


Figure 8.3: Chapter 8 mind map: dispersion, attenuation, and material energy follow from the same response model.

9 Polarization, Interference, and Wave Packets

Source map: Syllabus Units 6 and 8; Peatross and Ware Sections 6.1–6.6 and 7.1–7.5.

9.1 Polarization is a trajectory, not a propagation direction

For a wave along $+z$, let $\boldsymbol{E}_0 = A\hat{\boldsymbol{x}} + Be^{i\delta}\hat{\boldsymbol{y}}$, with $A, B \geq 0$. At a fixed position and after absorbing a common phase, $E_x = A\cos\omega t$ and $E_y = B\cos(\omega t - \delta)$. Eliminating time yields

$$\left(\frac{E_x}{A}\right)^2 + \left(\frac{E_y}{B}\right)^2 - 2\frac{E_x E_y}{AB}\cos\delta = \sin^2\delta.$$

The equation is understood by limits when A or B is zero. If $\delta = 0$ or π, the trajectory is a straight line. If $A = B$ and $\delta = \pm\pi/2$, it is a circle. Otherwise it is generally an ellipse.

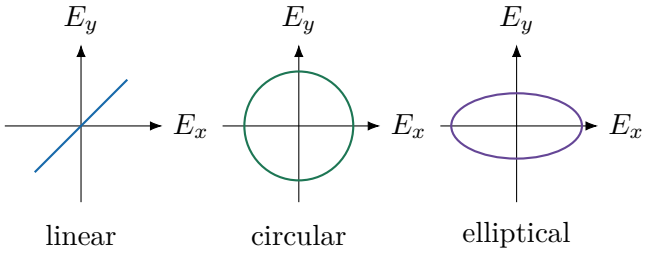


Figure 9.1: The electric-field tip at one position traces a line, circle, or ellipse over one period.

Definition: Jones vectors and the convention for rotation

A normalized Jones vector is $\boldsymbol{e} = (A, Be^{i\delta})^T/\sqrt{A^2 + B^2}$; multiplication by a common unit complex phase does not change its polarization state. With $e^{i(kz-\omega t)}$, the vector $(1, i)^T/\sqrt{2}$ rotates from $+x$ toward $+y$ at fixed $z = 0$ as time increases. Specify this trajectory and viewing direction before using right/left labels, since naming conventions differ. Peatross and Ware call this state left circular.

Worked Example: Determine a polarization and its intensity

For $\boldsymbol{E}_0 = E_a(2, \mathrm{i})^T$, $E_x = 2E_a \cos\omega t$, $E_y = E_a \sin\omega t$ at $z = 0$. Hence $E_x^2/(4E_a^2) + E_y^2/E_a^2 = 1$: an ellipse with axial ratio $1/2$. In a lossless nonmagnetic material, $I = (5/2)n\epsilon_0 c E_a^2$. The common scalar field scale is not the length of the major axis alone.

9.2 Projection and optional matrix tools

An ideal polarizer along real unit vector $\boldsymbol{p} = (\cos\theta, \sin\theta)^T$ produces $\boldsymbol{E}_{\text{out}} = \boldsymbol{p}\boldsymbol{p}^T\boldsymbol{E}_{\text{in}}$. For incident linear polarization at angle α,

$$I_{\text{out}} = I_{\text{in}} \cos^2(\alpha - \theta) \qquad \text{(Malus's law)}.$$

For any coherent polarization, use $I_{\text{out}}/I_{\text{in}} = |\boldsymbol{p}^\dagger \boldsymbol{e}|^2$. Unpolarized light transmits half its intensity through an ideal polarizer; a single Jones vector cannot represent that statistical mixture.

Supplementary Background: Jones matrices connect directly to linear algebra

The polarizer matrix is $P(\theta) = \boldsymbol{p}\boldsymbol{p}^T$, an orthogonal projector with $P^2 = P$. A lossless retarder aligned to x, y is $W = \mathrm{diag}(1, \mathrm{e}^{\mathrm{i}\delta_r})$. For axes rotated by θ, use $W(\theta) = R(\theta)WR(-\theta)$, where $R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$. The retardance is $\delta_r = (\omega/c)(n_{\text{slow}} - n_{\text{fast}})d$. Quarter- and half-wave plates have $\delta_r = \pi/2$ and π modulo 2π. Multiply device matrices in encounter order from right to left. Retarders preserve vector norm; polarizers generally do not.

Worked Example: A quarter-wave plate makes circular polarization

Send $\boldsymbol{e}_{\text{in}} = (1, 1)^T/\sqrt{2}$ through a plate with fast x axis and retardance $\pi/2$. Then $\boldsymbol{e}_{\text{out}} = (1, \mathrm{i})^T/\sqrt{2}$. Its norm remains one, so ideal transmission does not change intensity. Input $(1, 0)^T$ would stay linear: a quarter-wave plate does not make every input circular.

9.3 Derive the two-beam interference term

For coherent quasi-parallel waves in the same material, let local complex fields be $\boldsymbol{E}_1$ and $\boldsymbol{E}_2\mathrm{e}^{\mathrm{i}\Delta\phi}$. Using $I = C\boldsymbol{E}\cdot\boldsymbol{E}^*$ with $C = n\epsilon_0 c/2$ gives

$$I = I_1 + I_2 + 2C\,\mathrm{Re}(\boldsymbol{E}_1^* \cdot \boldsymbol{E}_2 \mathrm{e}^{\mathrm{i}\Delta\phi}).$$

If $\eta = \boldsymbol{e}_1^\dagger \boldsymbol{e}_2$ for normalized polarization vectors,

Key Result: Visibility and polarization overlap

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\,|\eta| \cos(\Delta\phi + \arg\eta),$$

$$I_{\text{max,min}} = I_1 + I_2 \pm 2\sqrt{I_1 I_2}|\eta|, \qquad \mathcal{V} = \frac{I_{\text{max}} - I_{\text{min}}}{I_{\text{max}} + I_{\text{min}}} = \frac{2\sqrt{I_1 I_2}|\eta|}{I_1 + I_2}.$$

Full visibility requires equal intensities and identical polarization up to a phase. Orthogonal Jones vectors give no cross term even when both beams are perfectly coherent.

Worked Example: Imbalance and polarization reduce contrast independently

If $I_1 = 9I_0$, $I_2 = I_0$, and the linear polarizations differ by $60°$, then $|\eta| = 1/2$. Therefore $I = 10I_0 + 3I_0 \cos\Delta\phi$, $I_{\text{max}} = 13I_0$, $I_{\text{min}} = 7I_0$, and $\mathcal{V} = 0.30$. Equalizing the intensities would raise visibility to 0.50, not to one, because the polarization mismatch remains.

9.4 Spatial fringes and optical path

For plane waves, $\Delta\phi = (\boldsymbol{k}_2 - \boldsymbol{k}_1)\cdot\boldsymbol{r} + \phi_0$. Fringes are planes of constant phase difference perpendicular to $\Delta\boldsymbol{k}$. In the symmetric crossing geometry with total crossing angle 2θ and common k, the transverse fringe spacing is

$$d = \frac{2\pi}{2k\sin\theta} = \frac{\lambda}{2\sin\theta}.$$

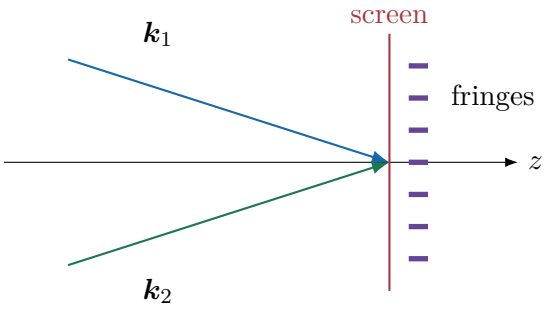


Figure 9.2: Two slightly crossing beams give a transverse sequence of phase-difference maxima.

In a nondispersive path of length ℓ, phase accumulation is $n\omega\ell/c = 2\pi n\ell/\lambda_0$. The optical path difference is $\Delta(\int n\,ds)$, with any reflection or device phase added separately.

Worked Example: Count fringe shifts from a path change

In a two-arm arrangement a mirror displacement Δx changes the round-trip path by $2\Delta x$. In air, the phase changes by $4\pi\Delta x/\lambda_0$, and $N = 2\Delta x/\lambda_0$ fringes pass a fixed detector. A $1.50\,\mu$m displacement at $\lambda_0 = 600\,$nm shifts five fringes. The two-pass geometry, not a mysterious extra phase rule, supplies the factor of two.

9.5 Unequal frequencies and the detector timescale

For nearby frequencies in the same polarization,

$$I(x,t) \simeq I_1 + I_2 + 2\sqrt{I_1 I_2}\cos(\Delta k x - \Delta\omega t + \phi_0),$$

after averaging over the rapid optical cycles but not over the beat period. The beat frequency is $|\Delta\omega|/(2\pi)$. A detector averaging over many beat periods sees only $I_1 + I_2$.

For a rectangular exposure of duration T_d centered at t_c, integrating the cosine gives a multiplier

$$\operatorname{sinc}(\Delta\omega T_d/2), \qquad \operatorname{sinc} x = \frac{\sin x}{x}.$$

Its absolute value reduces the measured contrast. Equal-frequency beams with a random relative phase also lose the averaged cross term. Frequency mismatch, polarization orthogonality, and phase randomness are distinct mechanisms.

Worked Example: A beat can exist but disappear from the measurement

Two equal-polarization beams differ by 1 kHz. A 1 ms exposure spans exactly one beat period, so $\Delta\omega T_d/2 = \pi$ and $\operatorname{sinc}\pi = 0$. A sufficiently fast photodiode resolves the oscillation, but that integrating detector records the constant sum. Neither conclusion contradicts coherent field superposition.

9.6 Group index, delay, and pulse spreading

In a transparent medium, $k(\omega) = n(\omega)\omega/c$. Therefore

$$v_g = \left(\frac{\mathrm{d}k}{\mathrm{d}\omega}\right)^{-1} = \frac{c}{n + \omega n'}, \qquad n_g = n + \omega n' = n - \lambda_0 \frac{\mathrm{d}n}{\mathrm{d}\lambda_0}.$$

The last equality follows from $\omega = 2\pi c/\lambda_0$. A slab of thickness d gives group delay $\tau_g = n_g d/c$, and extra delay relative to vacuum $(n_g - 1)d/c$. Phase delay nd/c is generally different.

For a narrowband amplitude $A(0,t) = A_0 \mathrm{e}^{-t^2/(2\tau_0^2)}$, define $k_2 = \mathrm{d}^2k/\mathrm{d}\omega^2$ and $\tau = t - k_0' z$. Its Gaussian spectrum, in the detuning $\Omega = \omega - \omega_c$, is proportional to $\mathrm{e}^{-\tau_0^2\Omega^2/2}$. After factoring out the carrier phase, second-order propagation therefore requires

$$A(z,t) = \frac{A_0\tau_0}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left[-\frac{\tau_0^2 - \mathrm{i}k_2 z}{2}\Omega^2 - \mathrm{i}\tau\Omega\right] \mathrm{d}\Omega.$$

Complete the square, or use $\int_{-\infty}^{\infty} \mathrm{e}^{-a\Omega^2 + b\Omega}\mathrm{d}\Omega = \sqrt{\pi/a}\,\mathrm{e}^{b^2/(4a)}$ for $\mathrm{Re}\, a > 0$. With $a = (\tau_0^2 - \mathrm{i}k_2 z)/2$ and $b = -\mathrm{i}\tau$, this gives

$$A(z,t) = \frac{A_0}{\sqrt{1 - \mathrm{i}k_2 z/\tau_0^2}} \exp\left[-\frac{\tau^2}{2\tau_0^2(1 - \mathrm{i}k_2 z/\tau_0^2)}\right].$$

The intensity width parameter becomes $\tau(z) = \tau_0\sqrt{1 + (k_2 z/\tau_0^2)^2}$. This is a supplementary quantitative extension of the syllabus group-velocity idea. It assumes negligible frequency-dependent loss and sufficiently small higher-order dispersion over the pulse spectrum.

Worked Example: Group delay from wavelength data

At $\lambda_0 = 500\,\mathrm{nm}$ let $n = 1.50$ and $\mathrm{d}n/\mathrm{d}\lambda_0 = -0.040\,\mu\mathrm{m}^{-1}$. Converting λ_0 to $0.500\,\mu\mathrm{m}$, $n_g = 1.520$. A $2.00\,\mathrm{cm}$ slab gives $\tau_g = 101.4\,\mathrm{ps}$ and an excess delay $34.7\,\mathrm{ps}$. Keeping wavelength and derivative units consistent is essential.

9.7 Electromagnetic standing waves: fields and flux

Superpose equal x-polarized waves moving in opposite z directions in vacuum. Their magnetic fields have opposite signs, so

$$E_x = 2E_a \cos kz \cos\omega t, \qquad B_y = \frac{2E_a}{c} \sin kz \sin\omega t.$$

Electric and magnetic antinodes are displaced by $\lambda/4$. The instantaneous flux is

$$S_z = \frac{E_a^2}{\mu_0 c} \sin(2kz)\sin(2\omega t), \qquad \langle S_z \rangle = 0.$$

Energy flows locally back and forth. The average total energy density is $\epsilon_0 E_a^2$, independent of z, even though the electric part alone is proportional to $\cos^2 kz$. The local electric-field-squared response used in optical interference is therefore not the same as net directed flux.

Error Check: Where an interference formula applies

The scalar formula $I \propto |E_1 + E_2|^2$ is a field-strength detector model for comparable, quasi-parallel waves. For arbitrary crossing angles, magnetic fields and the detector surface normal must be included through $\langle \boldsymbol{S} \rangle \cdot \hat{\boldsymbol{n}}$. For counterpropagating waves, compute net flux separately; it is not the standing-wave electric intensity pattern.

9.8 Derivation Workshop: Polarization Geometry and Detector Readout

9.8.1 Ellipse axes from a real symmetric matrix

At one position write $E_x = A\cos\theta$ and $E_y = B\cos(\theta - \delta)$. Then

$$\begin{pmatrix} E_x \\ E_y \end{pmatrix} = \underbrace{\begin{pmatrix} A & 0 \\ B\cos\delta & B\sin\delta \end{pmatrix}}_{S} \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}.$$

This maps the unit circle to an ellipse. The squared semiaxes are the eigenvalues of

$$SS^T = \begin{pmatrix} A^2 & AB\cos\delta \\ AB\cos\delta & B^2 \end{pmatrix},$$

namely

$$a_\pm^2 = \frac{A^2 + B^2 \pm \sqrt{(A^2 - B^2)^2 + 4A^2B^2\cos^2\delta}}{2}.$$

The orientation of an axis obeys $\tan 2\alpha = 2AB\cos\delta/(A^2 - B^2)$, but the corresponding eigenvalue is needed to decide whether it is major or minor. This is a direct application of real symmetric diagonalization in Algebra 2M. It is an optional calculation tool for the syllabus polarization topic, not an assumption that Jones vectors apply to unpolarized light.

Worked Example: A rotated ellipse that is not in quadrature

Let $A = 2E_a$, $B = E_a$, $\delta = \pi/3$. The axis matrix in units of E_a^2 is $\left(\begin{smallmatrix} 4 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. Therefore $a_\pm^2/E_a^2 = (5 \pm \sqrt{13})/2$. The axial ratio is $a_-/a_+ = \sqrt{(5 - \sqrt{13})/(5 + \sqrt{13})} = 0.4025$, and $\tan 2\alpha = 2/3$, so the major axis makes $\alpha = 16.85°$ with x. Trace and determinant check the eigenvalues: their sum is 5 and product is 3.

9.8.2 Visibility inversion: what can a contrast measurement tell us?

For fixed relative polarization overlap $|\eta|$, define $\rho = I_2/I_1$. Then $\mathcal{V} = 2\sqrt{\rho}|\eta|/(1 + \rho)$. If $|\eta| = 1$ and $s = \sqrt{\rho}$, the measured contrast obeys $\mathcal{V}s^2 - 2s + \mathcal{V} = 0$. For $0 < \mathcal{V} \le 1$,

$$s = \frac{1 \pm \sqrt{1 - \mathcal{V}^2}}{\mathcal{V}}, \qquad \rho = s^2.$$

The two ratios are reciprocal: contrast alone cannot tell which beam is stronger. If polarization overlap or phase stability is unknown, contrast alone cannot uniquely determine the intensity ratio either. At $\mathcal{V} = 0$, the finite-division formula must not be used; an absent beam, orthogonal polarizations, or averaged random phase can all explain zero contrast.

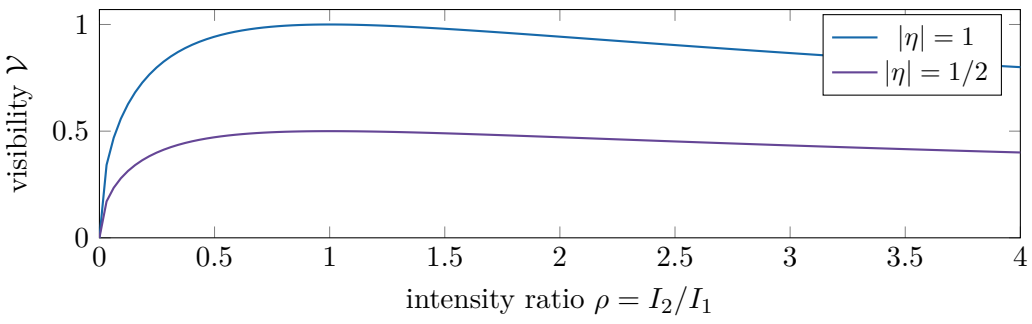


Figure 9.3: Power imbalance and polarization mismatch independently limit visibility. The maximum for fixed overlap occurs at equal intensities.

Worked Example: Two different experiments with the same visibility

Measured visibility is 0.80. For stable identical polarizations, $s = (1 \pm 0.60)/0.80 = 2$ or $1/2$, giving power ratios 4 or $1/4$. Alternatively, equal beam powers with overlap $|\eta| = 0.80$ produce exactly the same visibility. A contrast measurement must therefore be accompanied by individual power and polarization measurements to diagnose its cause.

9.8.3 Derive finite detector averaging rather than deleting a beat term

At fixed position, let $\Phi = \Delta k x + \phi_0$ and average $\cos(\Phi - \Delta\omega t)$ over an exposure centered on t_c. Set $t = t_c + s$. The odd sine term integrates to zero, while

$$\frac{1}{T_d}\int_{-T_d/2}^{T_d/2} \cos[\Phi - \Delta\omega(t_c + s)]\mathrm{d}s = \cos(\Phi - \Delta\omega t_c)\frac{2\sin(\Delta\omega T_d/2)}{\Delta\omega T_d}.$$

This equals the original beat cosine times $\mathrm{sinc}(\Delta\omega T_d/2)$. The limit as $\Delta\omega \to 0$ is one. A negative sinc factor shifts the observed fringe phase by π; its absolute value is the visibility reduction. The exposure, optical-cycle, beat, and phase-drift timescales must be distinguished.

Supplementary Background: Partial phase stability is not a new compulsory syllabus unit

If the relative phase fluctuates statistically, the interference cross term contains $\langle \mathrm{e}^{\mathrm{i}\phi} \rangle$ rather than a fixed $\mathrm{e}^{\mathrm{i}\phi}$. Its magnitude lies between zero and one, reducing contrast. More generally, mutual coherence can be represented by a normalized complex correlation when averaging over the detector process. This is a useful interpretation of lost visibility, but a full coherence theory is outside the explicitly supplied topic list.

Standard Workflow: A standard interference or polarization answer

1. Specify propagation directions, field phases, polarization vectors, and the measurement surface or local detector model.
2. Add complex vector fields before squaring. If an analyzer is present, project each field first.
3. Derive the cross term using the conjugate polarization overlap; then read extrema and visibility.
4. For unequal frequencies, retain the difference-frequency term until the exposure timescale is specified.
5. For a packet, differentiate $k(\omega)$ and state the bandwidth and loss assumptions; do not use ω/k for envelope motion.
6. Check identical/orthogonal polarizations, equal/unequal powers, and short/long exposure limits. Calculate net standing-wave flux separately.

9.9 Chapter Mind Map

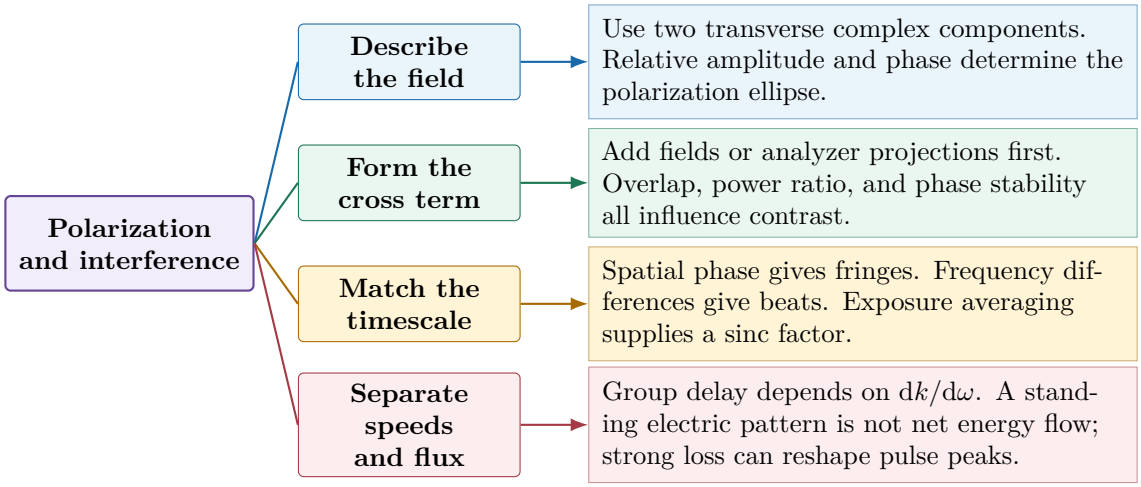


Figure 9.4: Chapter 9 mind map: the observed interference pattern depends on the field and on how it is measured.

10 Integrated Revision and Formula Map

10.1 One mathematical structure across the course

Model	Inertia	Restoring term	Response problem
single mass	m	k	$k - m\omega^2 - 2\mathrm{i}m\beta\omega$
coupled masses	M	K	$K - \omega^2 M - \mathrm{i}\omega C$
string	μ	$-\mathcal{T}\partial_x^2$	boundaries select k_n
EM field	ϵ	curl structure	$k^2 - \mu\epsilon\omega^2$
bound electron	m_e	$m_e\Omega_0^2$	$\Omega_0^2 - \omega^2 - \mathrm{i}\Gamma\omega$

Eigenvalue problems describe free motion. Inverting the driven operator describes forced motion. Orthogonality projects initial conditions and forcing into independent channels. Boundary conditions and constitutive laws decide which mathematical solutions are physical.

10.2 Derivation checklist by syllabus unit

Unit	Be able to derive, not just quote
1	Equilibrium curvature; initial-value motion in all damping regimes; phasor response; amplitude and power resonance; cycle-averaged power; linewidth and quality factor.
2	Spring extensions and stiffness matrix; symmetric/antisymmetric modes; mass orthogonality; beats; transverse coupling from tension and small slopes; forcing selection.
3	Monatomic dispersion; allowed k from boundaries; sound limit; binary determinant, both branches, gap and equal-mass limit; modal forcing.
4	String force balance; endpoint eigenfunctions; Fourier coefficients for shape and velocity; triangular pluck; modal energy; point-drive selection.

Unit	Be able to derive, not just quote
5	d'Alembert solution; energy continuity; impedance; interface and point-mass scattering; power conservation; v_p versus v_g.
6	Maxwell curl-curl derivation; transverse plane-wave constraints; E, B, H, k relations; Poynting conservation; average irradiance; attenuation and penetration lengths.
7	Electron displacement to polarization and susceptibility; real/imaginary parts; exact n, κ; heat production; dispersion and frequency limits.
8	Polarization ellipse; vector interference term; contrast; spatial fringe phase; finite-frequency beats and averaging; group delay; standing EM field and zero net average flux.

10.3 A concise equation index

Key Result: Dynamics and modes

$$\omega_0^2 = k/m, \quad \omega_d^2 = \omega_0^2 - \beta^2, \quad \omega_{\text{amp}}^2 = \omega_0^2 - 2\beta^2 \quad \text{when positive,}$$
$$Q = \omega_0/(2\beta), \quad \overline{P} = m\beta\omega^2 A^2, \quad K\boldsymbol{a} = \omega^2 M\boldsymbol{a},$$
$$\omega_{\text{mono}}^2 = 4K\sin^2(ka/2)/m, \quad v_g = \omega'(k), \quad v_p = \omega/k.$$

Key Result: Strings and interfaces

$$v = \sqrt{\mathcal{T}/\mu}, \quad Z = \sqrt{\mathcal{T}\mu}, \quad \omega_n = n\pi v/L,$$
$$P = -\mathcal{T}y_x y_t, \quad \langle P\rangle = Z\omega^2 A^2/2,$$
$$r_y = (Z_1 - Z_2)/(Z_1 + Z_2), \quad t_y = 2Z_1/(Z_1 + Z_2), \quad T_y = (Z_2/Z_1)|t_y|^2.$$

Key Result: Light, matter, and interference

$$\boldsymbol{B}_0 = \boldsymbol{k}\times\boldsymbol{E}_0/\omega, \quad \langle\boldsymbol{S}\rangle = \tfrac{1}{2}\,\mathrm{Re}(\boldsymbol{E}_0\times\boldsymbol{H}_0^*),$$
$$\chi = \frac{\Omega_p^2}{\Omega_0^2 - \omega^2 - \mathrm{i}\Gamma\omega}, \quad (n+\mathrm{i}\kappa)^2 = 1+\chi,$$
$$\alpha = 2\omega\kappa/c, \quad v_p = c/n, \quad v_g = c/(n+\omega n'),$$
$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\,\mathrm{Re}(\eta e^{\mathrm{i}\Delta\phi}), \quad \mathcal{V} = \frac{2\sqrt{I_1 I_2}|\eta|}{I_1 + I_2}.$$

10.4 An integrated worked application

Worked Example: Infer a material response from phase and transmission

A nonmagnetic uniform slab has internal wavelength 400 nm for vacuum wavelength 600 nm. Its internal intensity falls to e^{-1} after $d = 1.00$ mm. Determine the complex index, susceptibility, phase speed, and heating rate per volume at a point where $I = 100\,\mathrm{W\,m^{-2}}$.

Recognition and approach. Wavelength fixes the real index; transmission fixes α, hence κ; susceptibility follows by squaring, not by identifying χ with $n-1$.

First $n = \lambda_0/\lambda = 1.50$. Next $\alpha = 1/d = 1000\,\mathrm{m}^{-1}$, so $\kappa = \alpha\lambda_0/(4\pi) = 4.775 \times 10^{-5}$. Therefore

$$\chi' = n^2 - \kappa^2 - 1 \simeq 1.25, \qquad \chi'' = 2n\kappa = 1.432 \times 10^{-4}.$$

The phase speed is $c/1.5 = 1.999 \times 10^8\,\mathrm{m\,s}^{-1}$ and the local heating rate is $\alpha I = 1.00 \times 10^5\,\mathrm{W\,m}^{-3}$. Group speed cannot be determined from one measured index without frequency-derivative information. If the quoted transmission were external rather than internal, surface reflections would also need to be separated.

10.5 A complete, assessable answer

Standard Workflow: The six lines every long solution needs

1. Define coordinates, directions, amplitudes, and assumptions.
2. Derive the governing equation or boundary equations from physical laws.
3. Identify the method: initial-value solution, eigenmodes, Fourier projection, phasor inversion, or scattering match.
4. Show the decisive intermediate algebra or integral; separate singular or zero-parameter cases.
5. State the requested answer with units, phases, and subquestion labels.
6. Verify an initial condition, conservation law, sign, or limiting case that could expose an error.

Error Check: What cannot be predicted from these sources

The syllabus specifies required knowledge, not an examination distribution. The supplied optical textbook provides examples and exercises, but no past paper for this specific course. The practice set therefore labels all questions as original course-aligned training. Topic priority here means structural importance or conceptual difficulty, not a claimed statistical frequency in the school's examinations.

10.6 Source and production record

Subject sources inspected: `syllabus_waves.pdf`, two pages; Peatross and Ware, *Physics of Light and Optics*, 2015 edition, May 15, 2024 revision, 346 PDF pages. Only syllabus-relevant sections and explicitly marked short extensions are used. The supplied Algebra 1E V3.0 notes determine the visual style, not the subject scope. All diagrams are newly drawn vector illustrations. Equations use a single negative-time-exponential convention. The matching practice volume covers every syllabus unit with eight multipart problems and complete inline worked solutions.

10.7 A Standardized Decision Tree for Long Questions

Standard Workflow: Route A: Free or switched-drive motion

Trigger: initial position and velocity are given, or a force begins at a specified time.
Write: equilibrium and linear equation → characteristic roots → homogeneous motion → one particular response per drive frequency → constants fixed by the full initial data.
Output: a real motion with both quadratures, units, and any damping or resonance case stated. Check the two initial conditions and the differential equation. A steady phasor alone is not a switched-drive answer.

Standard Workflow: Route B: Modes and dispersion

Trigger: multiple coupled coordinates, a chain, a periodic cell, or a bounded string is specified.
Write: geometry and restoring terms → dynamic matrix or spatial operator → boundary-compatible trial functions → eigenvalues and eigenvectors → projections and reconstruction.
Output: all allowed modes, not just frequencies; include amplitude ratios and mode counts. Check mass weighting, zero modes, degeneracy, and any exceptional point where a ratio formula divides by zero.

Standard Workflow: Route C: Interfaces and energy transport

Trigger: a junction, localized barrier, slab, or a question about reflected/transmitted power is given.
Write: direction-correct incident/reflected/transmitted fields → continuity and force/field matching → complex amplitude coefficients → directed power coefficients.
Output: amplitude magnitudes and phases when requested, plus the correct impedance weighting for power. Check conservation and matched/fixed/free limits. State the reference planes if a propagation phase is included in t.

Standard Workflow: Route D: Material response and interference readout

Trigger: susceptibility, refractive index, attenuation, optical phase, visibility, or a detector exposure is given.
Write: fix the phasor convention → derive or reconstruct χ and $\widetilde{n}$ → identify the measured quantity → compute phase, heating, vector interference, or detector averaging as appropriate.
Output: distinguish phase speed, group delay, intensity attenuation, and amplitude attenuation. For interference, include polarization overlap. Check passivity, units, and information missing from an inverse problem.

10.8 A Complete Inverse-Response Example and Its Limitation

Worked Example: One complex susceptibility does not identify an entire material

Suppose $\chi(\omega_*) = u + iv$ with $v > 0$ is measured and a single Lorentz model is assumed. Then

$$\frac{1}{\chi} = \frac{u - iv}{u^2 + v^2} = \frac{\Omega_0^2 - \omega_*^2 - i\Gamma\omega_*}{\Omega_p^2}.$$

Equating parts gives

$$\Omega_0^2 = \omega_*^2 + \Omega_p^2 \frac{u}{u^2 + v^2}, \qquad \Gamma = \frac{\Omega_p^2 v}{\omega_*(u^2 + v^2)}.$$

There are two real observations but three real unknowns $\Omega_0, \Gamma, \Omega_p$. A one-parameter family of models remains: choose any positive Ω_p^2 yielding $\Omega_0^2 \geq 0$, and the formulas determine the other two. This explains why a single-frequency index and attenuation measurement does not determine group velocity.
If Ω_p is independently known, the inverse calculation becomes determinate. For example, in normalized units let $\omega_* = 1$, $\chi = 0.30 + 0.10i$, and $\Omega_p^2 = 0.20$. Then $u^2 + v^2 = 0.10$, $\Omega_0^2 = 1.60$, and $\Gamma = 0.20$. Substitution gives $\chi = 0.20/(0.60 - 0.20i) = 0.30 + 0.10i$, exactly checking the reconstruction. Without that independent density/strength measurement, these inferred material constants would not be unique.

10.9 A Final Coverage and Understanding Check

For each core syllabus unit, a complete revision should let you answer four different questions: What is the model? Why does its equation follow? How is a typical problem solved? What observation would show that the answer or approximation is wrong? The chapter workshops and maps answer these four questions explicitly. The V4.0 practice IDs remain useful, but the present notes do not claim that every conceivable examination variation is represented.

Error Check: Assumptions that must survive the final line

Small slope is not the same as small displacement; weak damping is not the same as weak coupling; weak absorption is not the same as weak susceptibility; a narrowband packet is not a single-frequency infinite wave; a decaying evanescent amplitude is not automatically heat production; and a fringe maximum is not automatically directed energy transport. Name the relevant condition rather than using the word "small" without a quantity.

10.10 Chapter Mind Map

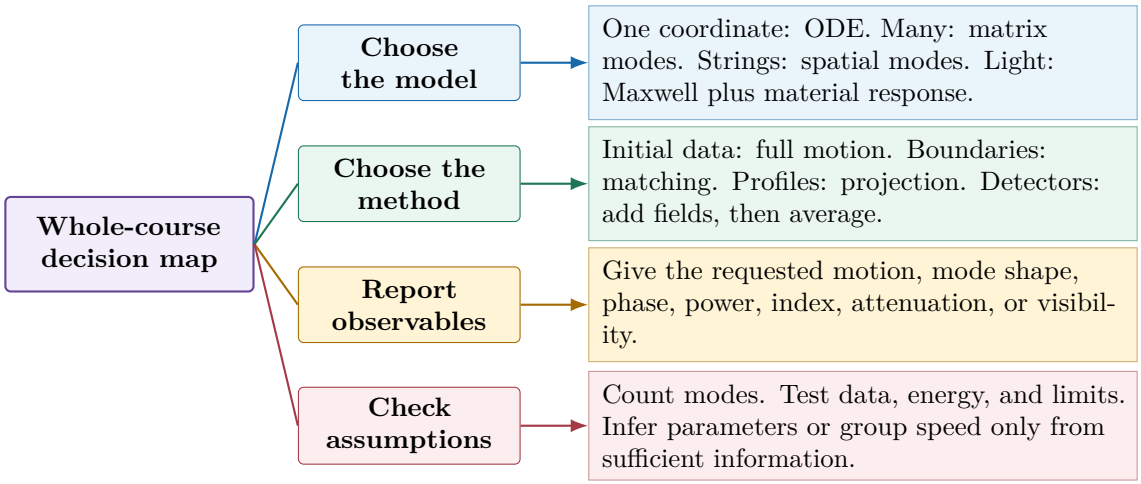


Figure 10.1: Chapter 10 mind map: recognize the problem structure before selecting a formula.